MIL-C-38488A(USAF)
1 September 1970

SUPERSEDING
MIL-C-38488(USAF)
10 February 1967

MILITARY SPECIFICATION

# COVERALL, FLYING, ANTI-EXPOSURE, QUICK DONNING CWU-16/P

## 1. SCOPE

1.1 This specification covers the requirements for one type and size of quick donning anti-exposure flying coverall designated Type CWU-16/P.

## 2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

| | |
|---|---|
| L-P-378 | Plastic Sheet And Strip (Polyolefin) |
| V-T-295 | Thread, Nylon |
| JJ-W-155 | Webbing, Textile, (Cotton, Elastic) |
| DDD-L-20 | Label, For Clothing, Equipage, And Tentage, (General Use) |

Military

| | |
|---|---|
| MIL-W-530 | Webbing, Textile, Cotton, General Purpose, Natural Or In Colors |
| MIL-B-543 | Buckles, Tongueless And Web Strap |
| MIL-T-5661 | Tape And Webbing, Textile, Cotton Reinforcing, Woven |
| MIL-C-7020 | Cloth, Parachute, Nylon |
| MIL-C-7219 | Cloth, Duck, Nylon, Parachute Packs |
| MIL-T-8363 | Tape And Webbing, Textile, Woven, Nylon |
| MIL-T-10884 | Fasteners, Snap |
| MIL-B-17757 | Boxes, Fiberboard, Corrugated (Modular Sizes) |
| MIL-C-19002 | Cloth, Coated, And Tape, Coated Cloth-Chloroprene On Nylon, Pneumatic Life Preserver |
| MIL-F-21840 | Fastener Tape, Hook And Pile, Nylon |
| MIL-C-23926 | Cloth, Coated And Laminated, Polychloroprene On Nylon, And Tape, Polychloroprene, Unsupported |



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STANDARDS

Federal

| | |
|---|---|
| FED-STD-191 | Textile Test Methods |
| FED-STD-595 | Colors |
| FED-STD-751 | Stitches, Seams, And Stitchings |

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures And Tables For Inspection By Attributes |
| MIL-STD-129 | Marking For Shipment And Storage |
| MIL-STD-831 | Test Reports, Preparation Of |
| MS27980 | Fasteners, Snap, Style 2 (Regular Wire Spring Clamp Type) |

DRAWING

Bureau of Naval Weapons

| | |
|---|---|
| 62A119D1 | Oral Inflation Assemblies - Inflatable Survival Equipment |

Air Force

| | |
|---|---|
| 66K1821 | Coveralls - Flying, Anti-Exposure, Quick Donning, Type CWU-16/P, Patterns For |

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. This specification provides for preproduction testing.

3.2 Materials and components. Materials and components that are not covered by specifications and drawings or are not specifically described herein shall be the best quality, the lightest weight practicable, and entirely suitable for the purpose intended. Except for the metallic parts and unless otherwise specified herein, the materials and the components used in the construction of the anti-exposure coverall shall have been manufactured not more than 12 months prior to the date of delivery of the coverall.

3.2.1 Coverall. The coverall shall be fabricated from cloth conforming to Type I of MIL-C-19002. The color of the cloth (crepe side) shall match sage green, USAF color shade No. 1511 (see 6.3).



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3.2.2 Neck and wrist seals. The neck and the wrist seals shall be fabricated from cloth conforming to Type I of MIL-C-23926 and to Figure 1. The color of the cloth shall be sage green, USAF color shade No. 1511 (see 6.3).

3.2.3 Outer sole and foot covering. The outer sole and the foot covering shall be fabricated from cloth conforming to Type III of MIL-C-7219. The color of the cloth shall approximately match the coverall fabric.

3.2.4 Hood, gloves, pockets, flaps, innersoles, patches, protective flap, and carrying case. The hood, the inner and the outer shells of the gloves, the pockets, the flaps, the innersoles, the patches, protective flap, and the carrying case shall be fabricated from cloth conforming to Type I of MIL-C-19002. The color of the cloth (crepe side) shall match sage green, USAF color shade No. 1511 (see 6.3).

* 3.2.5 Slide fastener. The water tight entrance slide fastener shall be B. F. Goodrich Co, Akron OH, Style 2430, or Talon Inc, Meadville PA, No 84, Type 1 (both ends closed and sealed) medium size, OEB, or equal. The slide fastener shall show no leakage when inspected as specified in 4.6.1. A thong, 3/8 inch wide by 10 inches long, nylon webbing strip conforming to Type I of MIL-T-8363 color sage green, USAF color shade No 1531, shall be looped through and securely knotted to the pull, approximately at the center of the webbing length. The ends of the thong shall be cut on an angle of approximately 60 degrees. The slide fastener shall conform to Table I.

TABLE I. Slide Fastener Dimensions

| Dimension | Requirements in Inches |
|---|---|
| Chain length | 37 +1/4, -1/2 |
| Opening in the rubber lip | 35 +1/4, -1/2 |
| Finished length of the thong | 4 ±1/4 |
| Width of slide fastener tape with the chain closed | 2 +1/16, -1/4 |
| Length of top end of slide-fastener tape extension | 1 minimum |
| Length of bottom end of slide-fastener tape extension | 1 minimum |



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* 3.2.6 Snap fasteners. The snap fasteners for the carrying case, the gloves, the pockets, and the glove adjustment straps shall conform to finish 2, style 2 of MIL-F-10884 and MS27980-1B, -6B, -7B, and -8B.

* 3.2.7 Lubrication of entrance slide fasteners. The rubber lips of the B. F. Goodrich entrance slide fastener shall be lubricated very lightly with Dow Corning 4 silicone compound (manufactured by Dow Corning Manufacturing Co, Midland MI) or a flourinated silicone and Teflon grease, M-125 (manufactured by American Oil Co, Whiting IN), or equal. The Talon slide fastener shall be lubricated by applying a light coating of paraffin wax to the chain halves.

3.2.8 Tape.

3.2.8.1 Seam tape. The seam tape for the gloves, the hood, and the coverall shall be fabricated from cloth conforming to Type I of MIL-C-7020 and polychloroprene coating compound conforming to MIL-C-19002. The seam tape shall consist of the layers specified in Table II and shall weigh 6.00 ±0.75 ounces per square yard. The calendered uncured polychloroprene gum coating shall be protected by a suitable liner which shall be capable of free separation without affecting the adhesion or cementability of the uncured gum coating. The edges of the seam tape shall be smooth, not pinked. The age of the tape shall be as specified in MIL-C-19002. The width of the tape used for binding the hood, the coverall seams, and the wrist edge of the gloves shall be 1 ±1/16 inch. The tape used in the gloves to interconnect the glove subassemblies and for covering the glove seams on the inner side of the glove shall be 3/4 ±1/16 inch. The color of all seam tapes shall be black or shall approximately match sage green, USAF color shade No 1511 (see 6.3). The seam tape shall not be dusted with talc, zinc stearate, or any other similar material.

TABLE II. Seam Tape Construction

| Layer No | Material | Weight (Ounces Per Square Yard) |
|---|---|---|
| 1 | Polychloroprene coating | 1.25 |
| 2 | Bias nylon cloth | 1.10 |
| 3 | Polychloroprene coating | 1.25 |
| 4 | Calendered uncured polychloroprene gum coating | 2.50 |



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3.2.8.2 Unsupported tape. The unsupported tape for the construction of the neck and the wrist seals shall conform to Type III, Class 1 or 2 of MIL-C-23926. The color shall be black or shall approximately match the color of the seal fabric.

* 3.2.8.3 Tie cords. The tie cords for attaching the gloves and the hood to the coverall pockets shall conform to Type II, 1/2 inch wide of MIL-T-5661 and shall match the color of the coverall fabric.

3.2.9 Webbing.

* 3.2.9.1 Wrist band. The inner wrist band of the gloves shall be fabricated from webbing conforming to Type I, Class 6 or 7, 1 inch wide, color optional of JJ-W-155.

* 3.2.9.2 Adjustment strap. The glove, ankle, and waist adjustment straps shall be fabricated from webbing conforming to Type II, Class 3, 3/4 inch wide of MIL-W-530 and shall match the color of the coverall fabric.

3.2.10 Thread. Thread conforming to Type I or II, Class 1 or 2, letter size E of V-T-295 shall be used for all sewing operations performed on the coverall. The dyed thread shall have at least good colorfastness to light and laundering and shall match the color of the coverall cloth.

3.2.11 Insulating material. The intermediate or insulating layer of the gloves shall be fabricated from urethane foam, color optional. The urethane foam shall have softness and flexibility suitable for apparel and shall be free from large holes or tears. The urethane foam shall conform to the requirements specified in Table III.

TABLE III. Insulating Material

| Characteristic | Requirement |
|---|---|
| Tensile strength, psi | 20 minimum |
| Elongation, percent | 250 minimum |
| Compression load at 25 percent deflection, psi | 0.4 minimum |
| Density, pounds per cubic feet | 2.7 maximum |
| Thickness, inch 1/ | 1/8 ±1/64 |

1/ The urethane foam may be one or two plies.



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3.2.12 Buckles. The buckles for all adjustment straps shall conform to Type II, Style 2, Class 1, size 3/4 inch of MIL-B-543.

3.2.13 Oral inflation assembly. The oral inflation assembly shall conform to Drawing 62A119D1, Assembly Part No 62A119D1-9.

* 3.2.14 Cement. The cement used in the fabrication of the coverall, the hood, and the gloves shall be chloroprene liquid cement. The color of the cement shall be neutral or shall approximately match the color of the material upon which the cement is used. When the solvent in the cement evaporates, a flexible film shall remain. The film shall have suitable tackiness, strength, aging, and working qualities for the intended purpose.

* 3.2.15 Fastener tape. The fastener tape for the hood shall conform to MIL-F-21840, Type II, 1 inch wide.

3.3 Design. The coverall shall be a one-piece garment. The coverall shall have two expandable type patch pockets, two waist adjustment straps, and attached boots, with adjustable ankle straps. A carrying case for stowing the coverall assembly, one pair of gloves with adjustable wrist straps, and a hood shall be furnished with each coverall. Each glove and each hood shall have a tie cord for attaching to the pockets of the coverall. The general appearance of the coverall shall conform to Figures 2 and 3.

3.4 Construction. The construction of the coverall assembly shall be accomplished by the use of sewn and cemented seams, as applicable. The details of the coverall assembly construction shall conform to Figures 4 through 19. The coated side of the coverall cloth shall be on the inner side of the coverall, the gloves, the hood, and the carrying case. The coated side of the neck and the wrist seal cloth shall be on the outer side of the seals.

3.4.1 Patches. Except for the seam areas and the neck and the wrist seals, the coverall may contain a maximum of five patches. The patches shall be applied to the coated side of the cloth and shall not exceed 1 inch in diameter. The patches shall be used to repair defective coated areas, a straight-line cut or tear that does not exceed 3/8 inch in length, or a triangular tear that does not exceed 3/8 inch from the base to apex nor a base width of 5/16 inch. The patch shall be centered approximately over the defective area; however, not less than 3/16 inch of patch shall extend past all parts of the defective area.

3.4.2 Cutting. The coverall, the boots, the neck and the wrist seals, the hood, the gloves, and the carrying case shall be cut in strict accordance with the furnished patterns (see 3.7) which show direction of cloth, size, shape, placement of pockets, and notches or markings for the proper assembly of all the parts.



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* 3.4.3 Cementing of the seams, glove seams, supported and unsupported tapes, patches, attachments, and sealing areas. The construction of the seams and the sealing areas and the cementing of the tapes, patches, and attachments shall be performed utilizing the technique and the precautions specified in 3.4.3.1 so that the adhesion of the assembled coverall including hood and gloves shall have developed the optimum bonding properties prior to their submission for inspection. All the seams, patches, sealed areas, tapes, and attachments shall be secured by the adhesive and all of the cementing shall be performed in accordance with the instructions specified by the adhesive manufacturer. The seam tape specified in 3.2.8.1 shall be cemented over all the stitched seams on the inner side of the coverall. The neck seal, where attached to the coverall, shall be covered on the inner and the outer sides with the seam tape specified in 3.2.8.1. The wrist seals, where attached to the coverall, shall be covered on the outer side with the seam tape specified in 3.2.8.1. The stitched seams of the inner and the outer shells of the gloves shall be covered with the seam tape specified in 3.2.8.1. All seam tapes shall be applied without tension and with not less than 3/16 inch of tape from either the raw edge or the folded edge of the stitched seam. All patches and attachments shall be applied to the coverall, the hood, and the gloves without tension. The adhesive shall be applied in an even line parallel to the edges of the tapes, seams, patches, and attachments. The adhesive shall extend a minimum of 1/8 inch or a maximum of 1/4 inch beyond the edges of the seam tapes, patches, or attachments. The ends of all the seam tapes shall be rounded. All supported tapes used on the coveralls, the hood, and the gloves shall overlap a minimum of 3/4 inch at the joining points.

3.4.3.1 Use of adhesive. In all the cementing operations, the surface to which the adhesive is to be applied shall be thoroughly cleaned with a suitable solvent so that the dusting materials (zinc stearate or talc) and any other surface contaminant is removed. The surface shall be clean prior to cementing. Care should be exercised to insure that the coating and the base undercloth are not damaged and the adhesion between both is not impaired in any way by prolonged exposure to the solvent. The solvent used shall evaporate completely prior to the application of the adhesive and shall leave no residue. Cemented areas shall be rolled out; all trapped air, channels, and wrinkles shall be removed. The adhesive shall be controlled to insure that old adhesive or adhesive which has partly or completely polymerized is not used. A fresh batch of the adhesive shall be used at least every 8 hours during the course of manufacture. Before containers for the adhesive are refilled, the containers shall be freed from congealed adhesive.

3.4.4 Stitches, seams, and stitchings. Except for the construction of the neck and the wrist seals and the protective flap, all the stitching used in the coverall shall conform to stitch Type 301 of FED-STD-751 and shall be 8 to 10 stitches per inch. The stitching used in the seam construction of the neck and the wrist seals shall be as specified in Figure 1. The stitching used in the construction of the protective flap shall be stitch Type 301 or 401, seam type SSa-1 of FED-STD-751 and 8 to 10 stitches per inch. Except for the hood, the neck and the wrist seals, the thong, and the tie cords, no raw or cut edges shall be visible. Each row of stitching shall be straight and parallel to the seam edges. Unless otherwise specified, all



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the top stitching shall be 1/4 inch from the edge. A tolerance of plus or minus 1/16 inch shall be allowed in the nominal distance of stitches from the edge. This nominal distance shall be maintained in any row within a tolerance of plus or minus 1/32 inch. The thread tension shall be maintained so that there shall not be any loose stitching and the lock shall be embedded in the center of the materials sewn together. Thread breaks, skips, and run-offs shall be overstitched not less than 1/2 inch. Ends of stitching that are not caught in other seams or stitching shall be backstitched not less than 1/2 inch.

3.4.4.1 Seam allowance. Unless otherwise specified, a seam allowance of approximately 3/8 inch shall be provided on all seams of the coverall, the gloves, and the carrying case.

3.4.4.2 Sewn eyelets. Each sewn eyelet shall contain a minimum of 30 stitches. The opening in the eyelet shall be 1/4 ±1/16 inch in diameter.

3.4.4.3 Bartack. The number of stitches per bartack shall be based proportionally on a 1/4-inch-long bartack containing a minimum of 14 stitches.

3.4.5 Neck and wrist seals. The neck and the wrist seals shall be constructed in accordance with Figure 1.

* 3.4.6 Entrance slide fastener with protective flap. The smooth outside edge area of the entrance slide-fastener tape shall be roughened and then cemented to the coated side of the coverall. The slide fastener chain shall be on the interior of coveralls when the Goodrich slide fastener is utilized and on the exterior when the Talon slide fastener is used. The entrance slide fastener shall close downwards, and the pull shall be on the outer side of garment. The entrance slide fastener shall have a protective flap conforming to the requirements of 3.4.6.1.

3.4.6.1 Protective flap. A protective flap shall be installed with either slide fastener. Fabrication and installation of the protective flaps shall be as follows:

(1) With the B. F. Goodrich, Style 2430 slide fastener, the protective flap shall be installed on the inside of the coverall over the slide fastener closure. The flap shall be fabricated from a section of coverall fabric that is cut 37 1/2 plus or minus 1/4 inches long by 6 inches wide. The fabric shall be folded lengthwise at the center, coated side to coated side, and stitched with a continuous single row of stitching around sides and ends. The stitching shall be 1/4 ± 1/16 inches from the folded and raw edges. A zig-zag type of stitching with the points 3 inches apart shall complete the flap construction. The stitching shall start at one end and continue the length of the flap, ending at the opposite end. The completed flap shall be attached to the coverall by cementing the side with the raw edges to the supported tape used on the slide fastener tape and coverall. The cemented area shall extend the full length of the flap and shall be 1 plus or minus 1/16 inch wide. The edge of the flap shall be



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even with the edge of the supported tape that is farthest from the center line of the fastener. The flap shall be positioned to cover all exposed metal parts of the fastener. (Seam Type SSa-1, stitch type 301 or 401, 8 to 10 stitches per inch.)

(2) With the Talon Inc No 84, Type 1, slide fastener, the protective flap shall be installed on the outside of the coverall over the slide fastener closure. The flap shall be fabricated from a section of coverall fabric that is cut 37 1/2 ± 1/4 inches long by 6 inches wide. The fabric shall be folded lengthwise at the center, coated side to coated side and stitched with a continuous single row of stitching around the sides and ends (Seam Type SSa-1). The stitching shall be 1/4 ± 1/16 inches from the folded and raw edges. A zig-zag type of stitching with the points 3 inches apart shall complete the flap. The stitching shall start at one end corner and continue the length of the flap ending at the corner in the opposite end. The flap shall be installed on the coverall by folding under the raw edges on the length of the flap 1/4 inches ± 1/16 inch and placing the flap over the entrance fastener with the turned under portion parallel to the length of the cutout on the left side of the coverall. The flap shall be fastened to the coverall with a single row of stitching the full length of the flap (Seam Type SSn-1). The row of stitching shall be not less than 5/8 nor more than 3/4 inches from the edge of the coverall cutout and 1/8 ± 1/16 inch from the folded edge of the turn under on the flap. The stitching shall not catch the fastener tape. The flap length shall be equally spaced over the cutout length. The ends of the flap shall not be stitched to the coverall but the lengthwise stitching shall be back stitched a minimum of 1/2 inch at each end. Seam tape shall be centered over stitching and fastener tape and cemented on inside of coverall.

* 3.4.7 Assembling completed coverall. All parts of each complete coverall assembly (coverall hood, and gloves) shall be covered with a minimum amount of zinc stearate or talc to prevent adherence of the adjacent cloth surfaces, tapes, and cemented areas during normal storage and transportation conditions. Care shall be exercised so that zinc stearate or talc shall not contaminate the entrance slide fastener of the coverall. The hood shall be completely evacuated of air. The oral inflation valve shall be locked in the closed position by the knurled ring. One end of the tie cord shall be securely attached to the sewn eyelet in the hood. The other end of the tie cord shall be securely attached to the sewn eyelet in the hem of the right-hand coverall pocket. The hood shall be folded in half and then rolled. The oral inflation tube and the valve shall be on the inner side of the roll. The hood shall be inserted into the right-hand pocket. The adjustment straps and the tie cords shall be snapped to the gloves. The free ends of the tie cords attached to the gloves shall be securely attached to the proper sewn eyelets in the hem of the pockets. The pocket flaps shall be snapped closed. The coverall slide fastener shall be closed to within 3-4 inches of the sealing block at the bottom of the coverall. The coverall assembly shall then be folded and rolled as follows:



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a. Place the coverall face down (see step No. 1 of Figure 20).

b. Place the sleeves across the back (see step No 2 of Figure 20).

c. Position the slide fastener on the underside of the coverall prior to rolling (see step No. 3 of Figure 20).

d. Position the coverall for rolling by folding the left side to the center back and then by folding the right side to the center back, slightly lapping the right pocket over the left one (see step No. 4 of Figure 21).

e. Roll the coverall so that the slide fastener will lay flat around the rolled coverall (see step No. 5 of Figure 21).

f. Insert the coverall into its carrying case (see step No. 6 of Figure 21). Snap the case closed.

3.5 Performance.

3.5.1 Entrance slide fastener leakage. When a pressure of 2 pounds per square inch (psi) is applied to the individual sections of the entrance slide fastener and it is immerged in water as specified herein, the entrance slide fastener shall not leak.

3.5.2 Porosity of the oral inflation tube. The oral inflation tube shall not leak nor be deformed and there shall not be any evidence of weakness of the tube when it is subjected to an internal pressure of 25 psi as specified herein.

3.5.3 Coverall leakage. When a pressure of 6 inches (water column) is applied as specified herein, the pressure in the coverall shall not drop below 5 inches (water column). The inflated coverall shall not show any evidence of constructional or material failure. All seams, sealed areas, and cemented attachments of the coverall shall remain perfectly intact and shall not show any sign of separation.

3.5.4 Hood leakage. When a pressure of 2 inches (water column) is applied to the hood as specified herein, the pressure in the hood shall not drop below 1.5 inches (water column). The inflated hood shall not show any evidence of constructional or material failure in any respect. The seams and the attachments shall remain perfectly intact and shall not show any indication of separation. There shall be no evidence that the air entering the compartment is hindered by the misplacement of the reinforcing patch on the back of the oral inflation valve or the opening in the oral inflation tube is blocked by excess adhesive, zinc stearate, or any other foreign matter. The oral inflation valve shall operate without difficulty, shall shut off securely, and shall not be blocked by any excess adhesive, talc, zinc stearate, or any other foreign matter.



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3.5.5 Glove water absorption. The water absorbed by the glove, when it is submerged in fresh water at 75 ±5°Fahrenheit (F) (24 ±3°Centigrade (C)) under a static head of 2 feet of water for a 2-hour period, shall not increase the weight of the glove more than 13 percent.

3.5.6 Breaking strength. The oral inflation tube shall have a minimum breaking strength of 50 pounds.

3.5.7 Seam tape adhesion. The minimum adhesive quality of the seam tapes shall not be less than 5 pounds per inch of width when tested as specified herein.

3.5.8 Strength of the cemented bond between the oral inflation valve and the oral inflation tube. A load of not less than 50 pounds shall be required to separate the cemented bond between the oral inflation valve and the oral inflation tube.

3.5.9 Marking ink (pH factor). The marking ink shall have an acidity (pH) of 5.0 to 8.5 and shall not contain any materials that are deleterious to the hood or the gloves.

3.5.10 Color matching. The color of the coverall assembly (coverall, hood, and gloves) shall match the approved standard shade under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin.

3.6 Marking. Each coverall shall have an identification label and an instruction label. In addition, each coverall shall be marked as specified in 3.6.5. Each glove and each hood shall be marked as specified herein with ink conforming to the requirements of 3.5.9. The marking on the hood and the gloves shall be legible and durable black letters and numerals and shall be thoroughly dry prior to packaging.

3.6.1 Identification label. The identification label shall conform to Type I, Class 1 of DDD-L-20 and shall be approximately 1-3/4 inches by 2-3/4 inches. The label is intended to be launderable. It shall be securely cemented or sewn on all four sides to the underside of the right pocket flap of the coverall. Except for the nomenclature, the legend of the label shall be in bold letters and numerals not less than 3/32 inch in height. The nomenclature shall be a minimum of 1/8 inch in height. The label shall contain the following information:

COVERALL, FLYING, ANTI-EXPOSURE, QUICK DONNING CWU-16/P
MIL-C-38488(USAF)
MANUFACTURER'S NAME
CONTRACT NUMBER
DATE OF MANUFACTURE (MONTH AND YEAR)



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3.6.2 Instruction label. The instruction label shall conform to Type I, Class 3 of DDD-L-20. The label is intended to be launderable. The instructions shall be printed in bold type that is 3/8 inch high. The label shall be securely cemented or sewn on all four sides to the carrying case as shown in Figures 18 and 19. The label shall contain the following information:

INSTRUCTIONS

1. OPEN COVERALL FASTENER UPWARD, AND PLACE BOTH FEET INTO COVERALL
2. PULL COVERALL TO THE WAIST: INSERT THE RIGHT ARM INTO RIGHT SLEEVE, PULLING COVERALL OVER THE SHOULDER.
3. PULL NECK SEAL OVER THE HEAD; INSERT THE LEFT ARM INTO LEFT SLEEVE.
4. CLOSE FASTENER DOWNWARD: USE SUFFICIENT FORCE TO SEAT THE PULL SECURELY INTO BOTTOM SEAL BLOCK.
5. REMOVE HOOD FROM POCKET AND DON.
6. POSITION, SNAP, AND INFLATE HOOD.

3.6.3 Hood marking. The inner side of each hood shall be marked in 1/4-inch letters and numerals with the following information:

HOOD
MIL-C-38488(USAF)
MANUFACTURER'S NAME
CONTRACT NUMBER
DATE OF MANUFACTURE 1/

1/The date of manufacture used shall be the month and the year the applicable coverall was manufactured.

3.6.4 Glove marking. The front wrist portion of each right-hand glove shall be marked with an "R" that is 1/2 inch in height. The front wrist portion of each left-hand glove shall be marked with an "L" that is 1/2 inch in height. The inner side of each glove, at the wrist section, shall be marked with the date of manufacture. The date of manufacture used shall be the month and the year the applicable coverall was manufactured.

3.6.5 Coverall marking. The front of each leg of the coverall shall be marked close beside and parallel with the rubber seal block of the slide fastener at the crotch area. The color of the marking ink shall conform to 13655, 23655, 17875, 27875, 33538, or 37875 of FED-STD-595. The instructions shall be placed on the coverall in a manner that will permit reading from above as the slide fastener is closed. The marking shall conform to Type IV of DDD-L-20 and shall be in commercial Gothic 18 point capital letters. The marking shall contain the following instruction: Seal pull tightly into rubber block.

3.7 Patterns. The coverall, the hood, the gloves, the boots, the carrying case, the neck seals, and the wrist seals shall conform to the patterns referenced on Drawing 66K1821. The patterns furnished are to be used as a guide for processing the manufacturer's working patterns. The patterns shall not be altered in any way without prior approval of the procuring activity.

3.8 Workmanship. The coveralls shall be manufactured in a thoroughly workmanlike manner. The coveralls shall be uniform in quality and shall be free from irregularities or any defects that could adversely affect performance, reliability, or durability. Because of the emergency and life support use of this coverall, the importance of providing a product of uniformly excellent quality cannot be overemphasized.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the coveralls shall be classified as follows:

a. Preproduction tests and inspections.

b. Quality conformance tests and inspections.

4.3 Preproduction testing.

4.3.1 Preproduction test samples. The preproduction test samples shall consist of two coverall assemblies (coverall, hood, and gloves) in their carrying cases that have been tested as specified in 4.3.2. The samples shall be representative of the construction, workmanship, components, and materials to be used during production. When a contractor is in continuous production of the items covered by this specification from contract to contract or within a 12-month period, submission of preproduction samples on the new contract may be waived at the discretion of the procuring activity (see 6.2). Approval of the preproduction test samples or the waiving of the preproduction test samples shall not preclude the requirement for the quality conformance tests and inspections.

4.3.2 Preproduction tests and inspections. The preproduction test samples specified in 4.3.1 shall be subjected to all of the tests and inspections described under 4.6. In addition standard commercial test methods shall be used to test the insulating material for the requirements specified in Table III.

4.3.3 Preproduction test report. When specified on DD Form 1423 and specified by the procuring activity (see 6.2), a preproduction test report in accordance with MIL-STD-831 shall be prepared and three copies shall be furnished with the test sample specified in 4.3.1.

4.4 Quality conformance tests and inspections.

4.4.1 Individual tests. Each coverall assembly (coverall, hood, and gloves) in its carrying case shall be subjected to and shall pass the following tests and inspections as described under 4.6:

a. Entrance slide fastener leakage.

b. Porosity of the oral inflation tube.



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c. Coverall leakage.

d. Hood leakage.

e. Glove water absorption.

f. Visual examination for critical and major defects.

4.4.2 Sampling tests.

4.4.2.1 Plan A. The inspection lot size for plan A shall be expressed in units of one coverall assembly (coverall, hood, and gloves) with its carrying case and made under essentially the same conditions and from the same materials and components. The sample unit shall be one coverall assembly (coverall, hood, and gloves) in its carrying case. The inspection level shall be II of MIL-STD-105. The acceptable quality level shall be 6.5 defects per 100 units. Samples selected in accordance with plan A shall be subjected to the visual examination for minor defects described in 4.6.7.

4.4.2.2 Plan B. The inspection lot size for plan B shall be expressed in units of one oral inflation tube and shall consist of all the oral inflation tubes received by the coverall manufacturer at one time. The sample unit shall be one oral inflation tube. The inspection level shall be S-2 of MIL-STD-105. The sample size shall be based only on the applicable sample code letter corresponding to the specified inspection level of MIL-STD-105. The acceptance number shall be zero, and the rejection number shall be one. The samples selected in accordance with plan B shall be subjected to the breaking strength as described in 4.6.8.

4.4.2.3 Plan C. The samples for plan C shall consist of sample seams. A sample seam shall be fabricated every 8 hours during production for each batch of cement used in the construction of the coverall assembly (coverall, hood, and gloves). The sample seams shall be subjected to and shall pass the test specified in 4.6.9.

4.4.2.4 Sampling plan D. The inspection level for plan D shall be S-2 of MIL-STD-105. The total acceptable quality level shall be 4.0 percent defective. The sample unit shall be one shipping container containing coverall assemblies (coverall, hood, and gloves in their carrying case) fully prepared for delivery with the exception that it need not be sealed. The inspection lot size shall be expressed in units of fully prepared shipping containers containing coverall assemblies (coverall, hood, and gloves in their carrying case) fully prepared for delivery from essentially the same materials and components. Each of the fully prepared shipping containers containing coverall assemblies (coverall, hood, and gloves in their carrying case) selected as a sample unit from the lot shall be examined as specified in 4.6.10.

4.5 Test and inspection conditions. Unless otherwise specified, all inspections and tests required by this specification shall be conducted at an atmospheric pressure of 28 to 32 inches of mercury, at a temperature



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of 77 ±18°F (25 ±10°C), and at a relative humidity of 80 percent or less. When tests or inspections are made with the atmospheric pressure or temperature substantially different from these values, proper correction shall be made for the change in instrument reading.

4.6 Test and inspection methods.

4.6.1 Entrance slide fastener leakage. Before the entrance slide fastener is incorporated into the coverall, it shall be inspected for leakage. The slide fastener shall be inserted into a suitable jig and oil and liquid water free air pressure of 2 psi shall be applied to the individual sections of the slide fastener. While the pressure is applied to the lip side of the slide fastener, the chain side of the slide fastener shall be immersed in water for 30 seconds. There shall be no evidence of leakage through the slide-fastener chain. Each slide fastener conforming to the no leakage requirement shall have a white letter "W" stamped on its tape.

4.6.2 Porosity of the oral inflation tube. Before the oral inflation valve is cemented within the oral inflation tube and to the hood, it shall be tested for porosity. While submerged in water, the oral inflation tube shall be subjected to an internal oil and liquid water free air pressure of 25 psi for a minimum of 20 seconds. While the pressure is applied, the inflation tube shall be examined. Any evidence of leakage caused by porosity or any evidence of weakening of the tube as revealed by excessive deformation in any portion of the inflation tube shall be cause for rejection.

4.6.3 Coverall leakage. A minimum of 24 hours after the last cementing operation in the coverall assembly, the coverall shall be turned inside out. The wrists and the neck shall be sealed, and the coverall shall be inflated with oil and liquid water free air to a pressure of 6 inches (water column). The air supply shall then be securely shut off. After a minimum period of 5 minutes, the pressure shall be measured. The pressure shall be not less than 5 inches (water column). While inflated, the coverall shall be examined. The coverall shall not show any evidence of constructional or material failure. All the seams, sealed areas, and cemented attachments of the coverall shall remain perfectly intact and shall not show any sign of separation.

4.6.4 Hood leakage. A minimum of 24 hours after the last cementing operation in the hood, the hood shall be inflated with oil and liquid water free air to a pressure of 2 inches (water column). The air supply shall be securely shut off. After a minimum of 5 minutes, the pressure shall be measured. The pressure shall be not less than 1.5 inches (water column). While inflated, the hood shall be examined for defects. There shall not be any evidence of constructional or material failure in any respect. The cemented seam and the attachments shall remain perfectly intact and shall not show any sign of separation. The oil and liquid water free air shall be introduced into the hood through the oral inflation valve and the tube to assure that there is no blockage or hindrance of the air at the back of the oral inflation valve or at the base of the oral inflation tube caused by excess adhesive, talc, zinc stearate, or any other foreign matter or by



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misplacement of the reinforcing patch. The cemented bond between the oral inflation valve and the tube and between the hood and the oral inflation tube shall remain perfectly intact during and after this inspection. The air from the compartment shall be discharged through the oral inflation assembly. The oral inflation valve shall operate without difficulty, shall shut off securely, and shall not be blocked by any excess adhesive, talc, zinc stearate, or any other foreign matter.

4.6.5 Glove water absorption. A glove, without the tie cord and the adjustment strap, shall be weighed on a scale or balance calibrated to weigh in increments of 0.1 ounce. The weighed glove shall then be submerged in fresh water at 75 ±5°F (24 ±3°C) under a static head of 2 feet of water for a 2-hour period. Immediately after the 2-hour period, the glove shall be removed from the water. The inner and the outer shells shall be wiped dry, and the gloves shall be reweighed. The weight increase of the glove shall be not greater than 13 percent. The glove shall then be dried without the application of heat, and the adjustment strap and the tie cord shall be snapped to the glove.

4.6.6 Visual examination for critical and major defects. Each coverall assembly (coverall, hood, and gloves) with its carrying case shall be examined visually for the critical and the major defects listed in table IV.

4.6.7 Visual examination for minor defects. The coverall asssembly (coverall, hood, and gloves) with its carrying case shall be examined visually for the minor defects specified in table IV.

4.6.8 Breaking strength. The oral inflation tube base shall be folded longitudinally so that the round edges coincide and the fold line is perpendicular to the main axis of the tube. The folded base shall then be placed in the lower jaw of a suitable inspection apparatus. The tube portion shall be placed in the upper jaw. The distance between the jaws at the start of the inspection shall be a minimum of 3 inches. The rate of jaw separation under no load shall be 12 ±1/2 inches per minute. The breaking strength of the oral inflation tube shall be not less than 50 pounds.

* 4.6.9 Seam tape adhesion. The adhesion of the seam tape to the sample seams shall be tested as specified in Method 5960 of FED-STD-191. A strip of the required seam tape, 8 inches in length, shall be cemented over a sample coverall seam in the same manner used for the coverall. Two inches of the seam tape shall be left uncemented. The completed sample seam shall measure 3 inches wide by 8 inches long and shall be allowed to cure prior to testing.



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TABLE IV. Classification of Defects

| DEFECT | CRITICAL | MAJOR | MINOR |
|---|---|---|---|
| GENERAL | | | |
| Hole, scissor or knife cut, tear, burn, or weakening defect such as a smash, multiple float, loose slub, needle chew, and defective coating areas (not patched) resulting from the coating process or caused by the manufacturing procedures | X | | |
| Color of any component not as specified | | | X |
| Coated side of the coverall cloth on the outer side | | X | |
| Thread ends not trimmed or loose thread ends not removed throughout the coverall | | | X |
| Any spot or stain | | | X |
| Any pocket or flap out of horizontal or vertical alignment by more than 1/4 inch | | | X |
| Any patch in the seams or the neck and the wrist seals, any coverall containing six or more patches, or any circular patch greater than 1 inch in diameter | | X | |
| GLOVE SEAM, SUPPORTED, AND UNSUPPORTED TAPES | | | |
| Neck-seal attachment not covered on the inner and the outer sides or not covered with the specified tape | X | | |
| Wrist-seal attachment not covered on the outer side or not covered with the specified tape | X | | |
| Slide-fastener attachment not covered on the inner side or not covered with the specified tape | X | | |
| Any supported tapes that overlap on themselves less than 3/4 inch | | X | |
| Any damaged tape edges greater than 1/16 inch from outside edge | | | X |



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TABLE IV. Classification of Defects (Cont)

| DEFECT | CRITICAL | MAJOR | MINOR |
|---|---|---|---|
| METALLIC COMPONENTS | | | |
| Surface rough; misaligned; or containing nicks, burrs, sharp edges, or any other flaws | X | | |
| Any component loose or otherwise not securely retained | | X | |
| Any functioning part that works with difficulty | X | | |
| One missing, loose, mismated, or otherwise defective snap fastener 1/ | | 2/ | |
| More than one missing, loose, mismated, or otherwise defective snap fastener 1/ | | X | |
| Snap-fastener section not properly positioned in relationship to corresponding mating section | | | X |
| SLIDE FASTENER 3/ | | | |
| Any part of the assembly bent, broken or otherwise defective | X | | |
| Not lubricated, heavily lubricated, or lubricated with material other than the specified silicone compound | | X | |
| Not closing downward | | X | |
| Length of slide fastener not as specified | | X | |
| COMPONENTS AND ASSEMBLY | | | |
| Any defective component or defect of assembly not herein classified | X | | |
| Any component part or required operation omitted or any operation improperly performed not herein classified | X | | |
| Any component part not cut in accordance with the patterns | | X | |



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TABLE IV. Classification of Defects (Cont)

| DEFECT | CRITICAL | MAJOR | MINOR |
|---|---|---|---|
| CEMENTING | | | |
| Cement on cloth surfaces along the seam tape or attachment edges 1/16 inch or less or in excess of 1/4 inch | | | X |
| Any seam tape not located as specified over any seam | | | X |
| SEAMS AND STITCHING | | | |
| Any seam or attachment of any component twisted, puckered, pleated, or caught in any unrelated operation or stitching | | X | |
| Ends of stitching not securely back-stitched for at least 1/2 inch when not caught in other seams | | X | |
| Thread breaks not securely overstitched at least 1 inch | | | X |
| Any stitching irregular or unevenly gaged (to be scored only when the condition exists on the major portion of the seam) | | | X |
| Not specified seam or stitch type | | X | |
| Loose tension, resulting in a loose seam or tight tension resulting in breaking of stitches when normal pull is applied 4/ | | X | |
| Stitches per inch one or two stitches more or less than specified (to be scored only when existing on the major portion of the seam) | | | X |
| Stitches per inch more than two stitches more or less than specified (to be scored only when existing on the major portion of the seam) | | X | |
| LABELS AND MARKINGS | | | |
| Identification label missing, incorrect, incomplete, illegible, or insecurely attached | | | X |



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TABLE IV. Classification of Defects (Cont)

| DEFECT | CRITICAL | MAJOR | MINOR |
|---|---|---|---|
| Instruction label missing, incorrect, incomplete, illegible, improperly located, or insecurely attached | | X | |
| Hood or glove marking missing, incomplete, incorrect, illegible, or located on the outer side of the hood | | | X |

1/The snap fasteners shall be checked for proper function and attachment by snapping closed and unsnapping each snap fastener at least three times.

2/If the defect is not in the gloves, or the glove adjustment straps, it shall be classified as minor defect. If the defect is in the gloves, or the glove adjustment straps it shall be classified as a major defect.

3/The slide fastener shall be checked for proper function by opening and closing the slide fastener at least three times along its full length.

4/Puckering is evidence of tight tension. When puckering is evident, the seam shall be inspected by exerting normal pull in the lengthwise direction of the seam or stitching.



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4.6.10 Examination of preparation for delivery. The fully prepared shipping container containing the coverall assemblies (coverall, hood, and gloves in their carrying case) shall be visually examined for the defects listed in Table V to determine if packaging, packing, and marking conforms to the specified requirements.

TABLE V. Defects In Preparation For Delivery

| Item | Defect |
|---|---|
| Carrying case | Not individually sealed in the polyethylene bag |
| | Case not snapped closed |
| Exterior and interior marking | Missing, incorrect, incomplete, or illegible |
| | Improper size, location, sequence, or method of application |
| Materials | Any nonconforming component |
| | Component missing, damaged, or otherwise defective |
| Workmanship | Inadequate application of the components such as incomplete closure of the container flaps, polyethylene bag, or loose strapping |
| Content | Number per container is more or less than required |

4.6.11 Strength of the cemented bond between the oral inflation valve and the oral inflation tube. The strength of the cemented bond between the oral inflation valve and the oral inflation tube shall be determined by placing the valve in the upper jaw and the tube in the lower jaw of a suitable inspection apparatus. The distance between the jaws at the start of the inspection shall be not less than 3 inches. The rate of jaw separation under no load shall be 12 ±1/2 inches per minute. The load required to separate the oral inflation valve from the oral inflation tube shall be not less than 50 pounds.



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4.6.12 Marking ink (pH factor). Forty milliliters of distilled water and forty grams of the marking ink shall be poured into a 4-ounce glass bottle. A nonmetallic stopper shall be inserted into the bottle. The mixture shall be allowed to stand at room temperature for 1/2 hour with occasional agitation. The bottle and its contents shall then be allowed to stand without further agitation for an additional 1/2 hours. The temperature shall be regulated to 77 ±2°F (25 ±1°C), and the pH shall be measured with a potentiometric apparatus. The acidity of the marking ink shall be between 5.0 and 8.5.

4.6.13 Color matching. The color of the coverall assembly (coverall, hood, and gloves) shall be compared with the approved standard shade under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin. The color of the coverall assembly shall match the approved standard shade.

## 5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. The coverall assembly (coverall, hood, and gloves in their carrying case) prepared as specified in 3.4.7 shall be packaged within a snug fitting bag. The bag shall be made of 0.0015-inch-gage polyethylene conforming to Type I; grade A, B, or C; finish 1 of L-P-378. A paper label (see 5.3.1) shall be inserted into the bag, and the bag shall be heat sealed. Prior to and during the sealing operation, excess air shall be exhausted to the extent necessary to permit packing of the required quantity in the shipping container.

5.1.2 Level C. The coverall assembly (coverall, hood, and gloves in their carrying case) prepared as specified in 3.4.7 shall be packaged to afford the minimum degree of protection necessary to prevent deterioration or damage during shipment under normal environmental conditions and commercial modes of transportation.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Six coverall assemblies in their carrying cases packaged as specified in 5.1.1 and placed in two equal tiers shall be packed within an export type of shipping container conforming to size 3A, class weather-resistant of MIL-B-17757. The container and the liner shall be constructed, closed, and reinforced as specified in MIL-B-17757.

5.2.2 Level B. Six coverall assemblies in their carrying cases packaged as specified in 5.1.1 shall be packed in two equal tiers within a domestic type fiberboard shipping container conforming to size 3A of MIL-B-17757. The container and the liner shall be constructed, closed, and reinforced as specified in MIL-B-17757.

5.2.3 Level C. The coverall assemblies in their carrying cases and packaged as specified in 5.1.2 shall be packed within exterior type shipping containers in a manner that shall insure transportation at the lowest rate to the point of delivery. The shipment shall conform to the minimum requirements of the rules and regulations applicable to the mode of transportation selected.



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5.3 Marking. The exterior containers shall be marked in accordance with MIL-STD-129. The marking shall include the following:

COVERALL, FLYING, ANTI-EXPOSURE QUICK DONNING CWU-16/P
MIL-C-38488(USAF)
MANUFACTURER'S NAME
CONTRACT NUMBER
DATE OF MANUFACTURE (MONTH AND YEAR)

5.3.1 Additional marking. Prior to heat sealing the polyethylene bag (see 5.1.1), a white paper label containing the nomenclature of the item in black letters and numerals shall be placed inside each bag so that it can be read through the bag.

6. NOTES

6.1 Intended use. The coverall covered by this specification is intended for quick donning (approximately 1 minute) by crew members in heavy bomber or cargo aircraft prior to emergency ditching of the aircraft at sea. After ditching of the aircraft, the coverall is intended to protect the wearer from exposure while swimming in cold water and from exposure to wind, spray, and rain while marooned in a life raft.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Waiver of preproduction test samples (see 4.3.1).

c. Preproduction test report requirement (see 4.3.3).

d. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Color samples. Color samples may be obtained from the procuring activity or as directed by the contracting officer.

* 6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 82

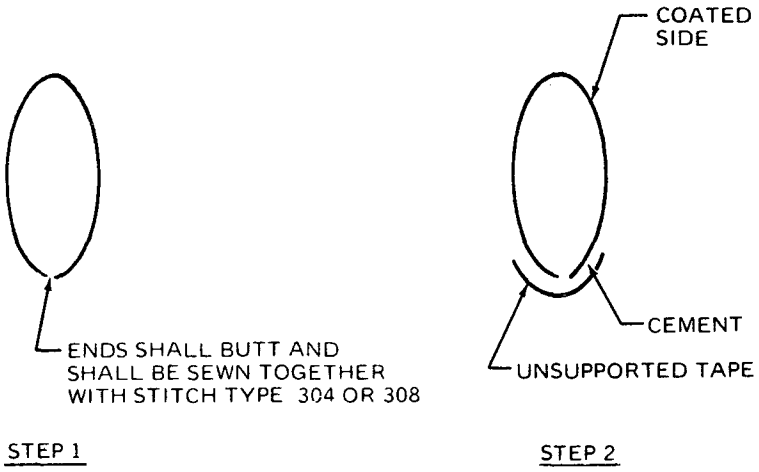
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Preparing Activity:
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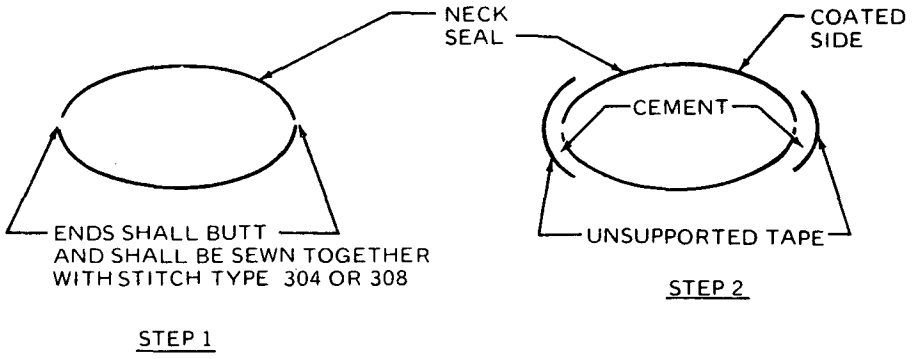
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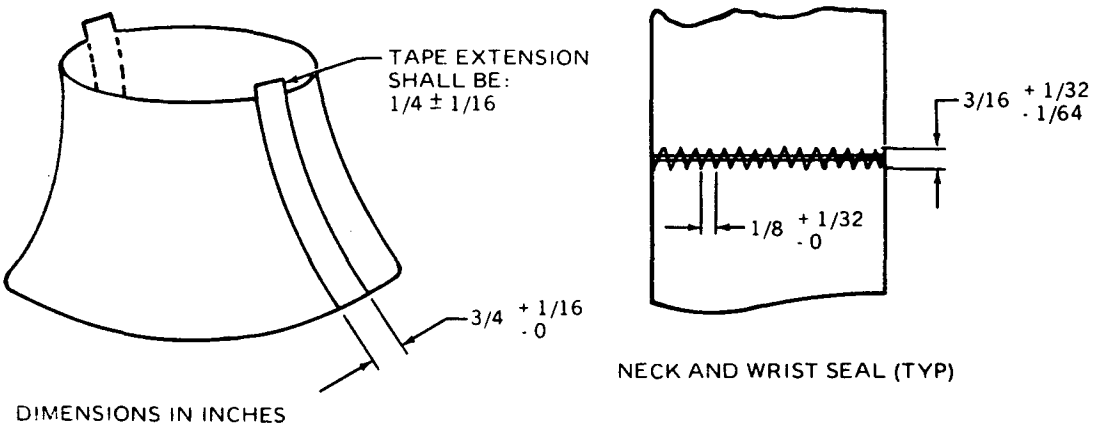


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CONSTRUCTION OF WRIST SEAL



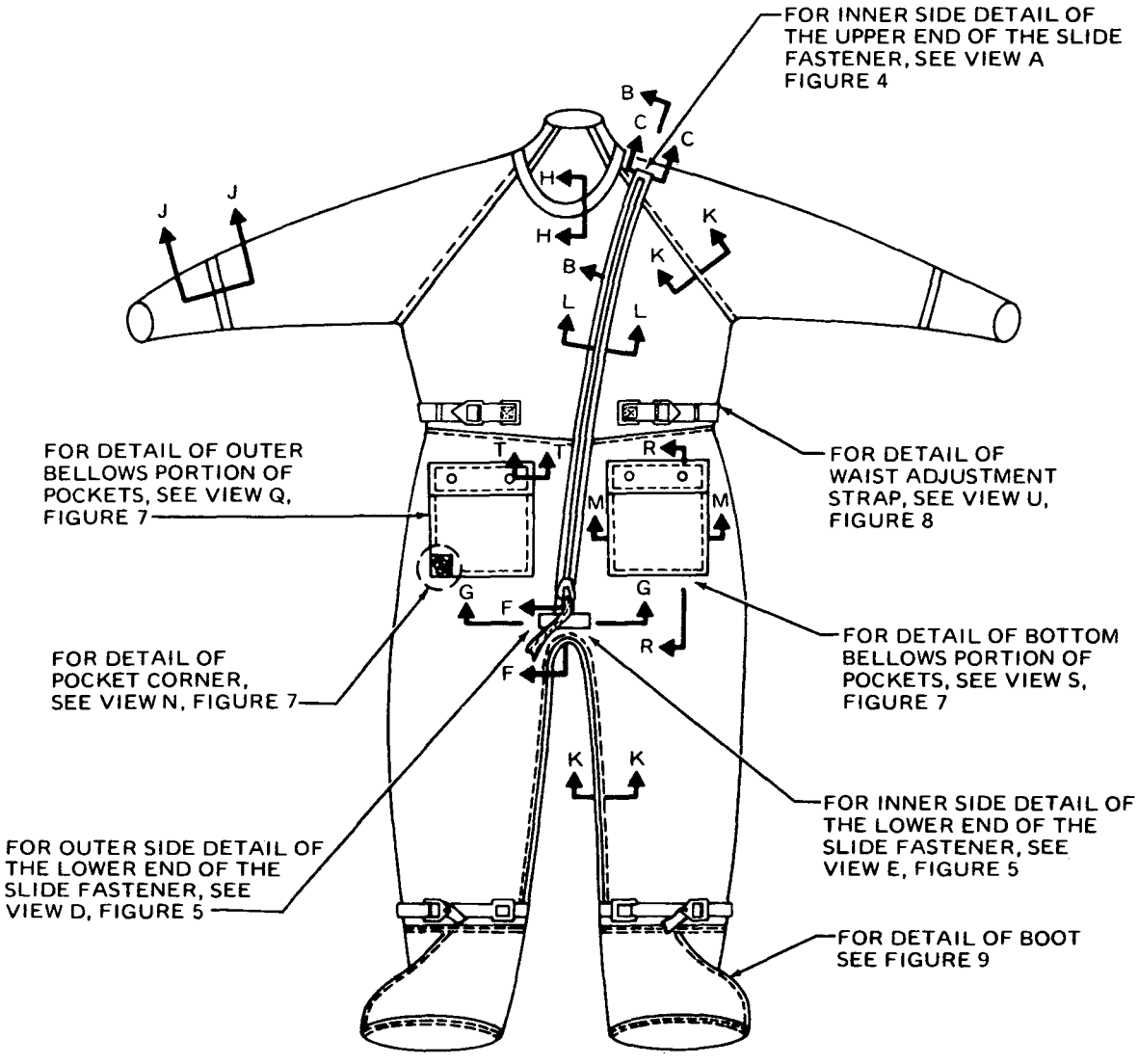




CONSTRUCTION OF NECK SEAL

FIGURE 1. WRIST AND NECK SEAL DETAILS



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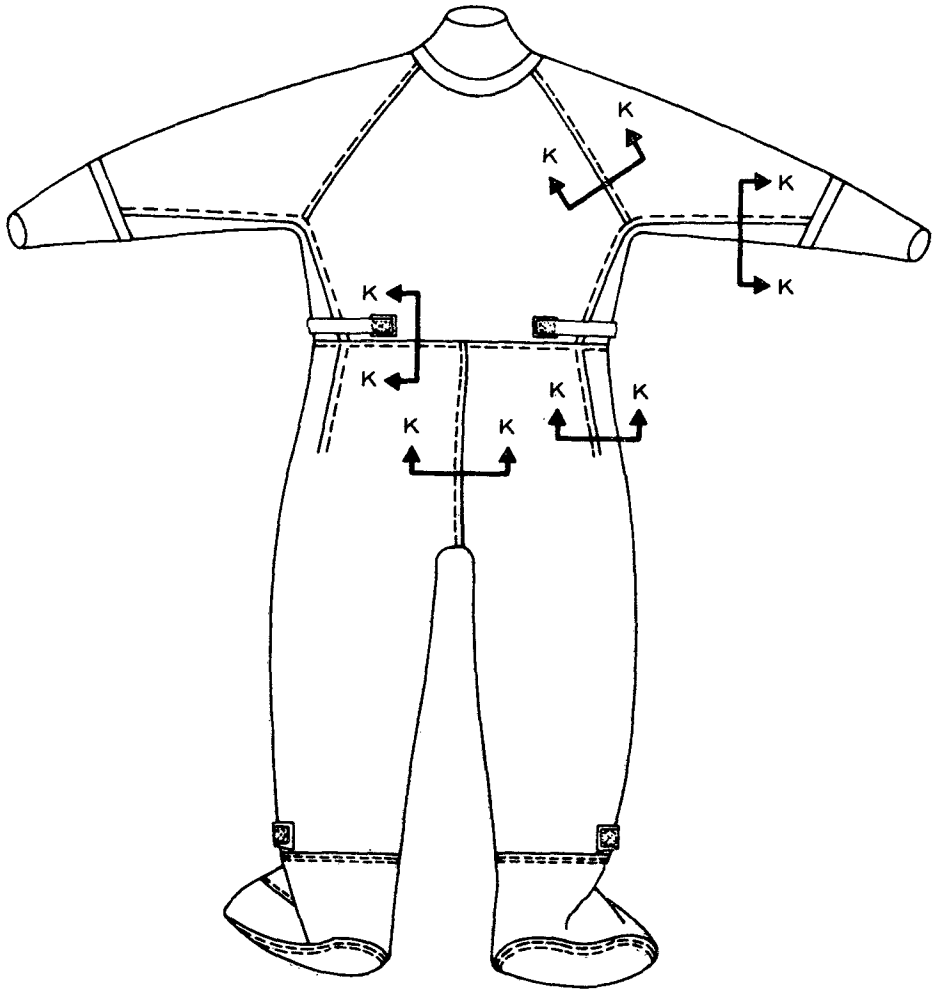



FOR SECTIONS B–B AND C–C, SEE FIGURE 4
FOR SECTIONS B–B AND C–C, SEE FIGURE 4
FOR SECTIONS F–F AND G–G, SEE FIGURE 5
FOR SECTIONS H–H, J–J, K–K, L–L, AND M–M, SEE FIGURE 6
FOR SECTIONS R–R AND T–T, SEE FIGURE 7

FIGURE 2. FRONT



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FOR SECTION K—K, SEE FIGURE 6.

FIGURE 3. BACK



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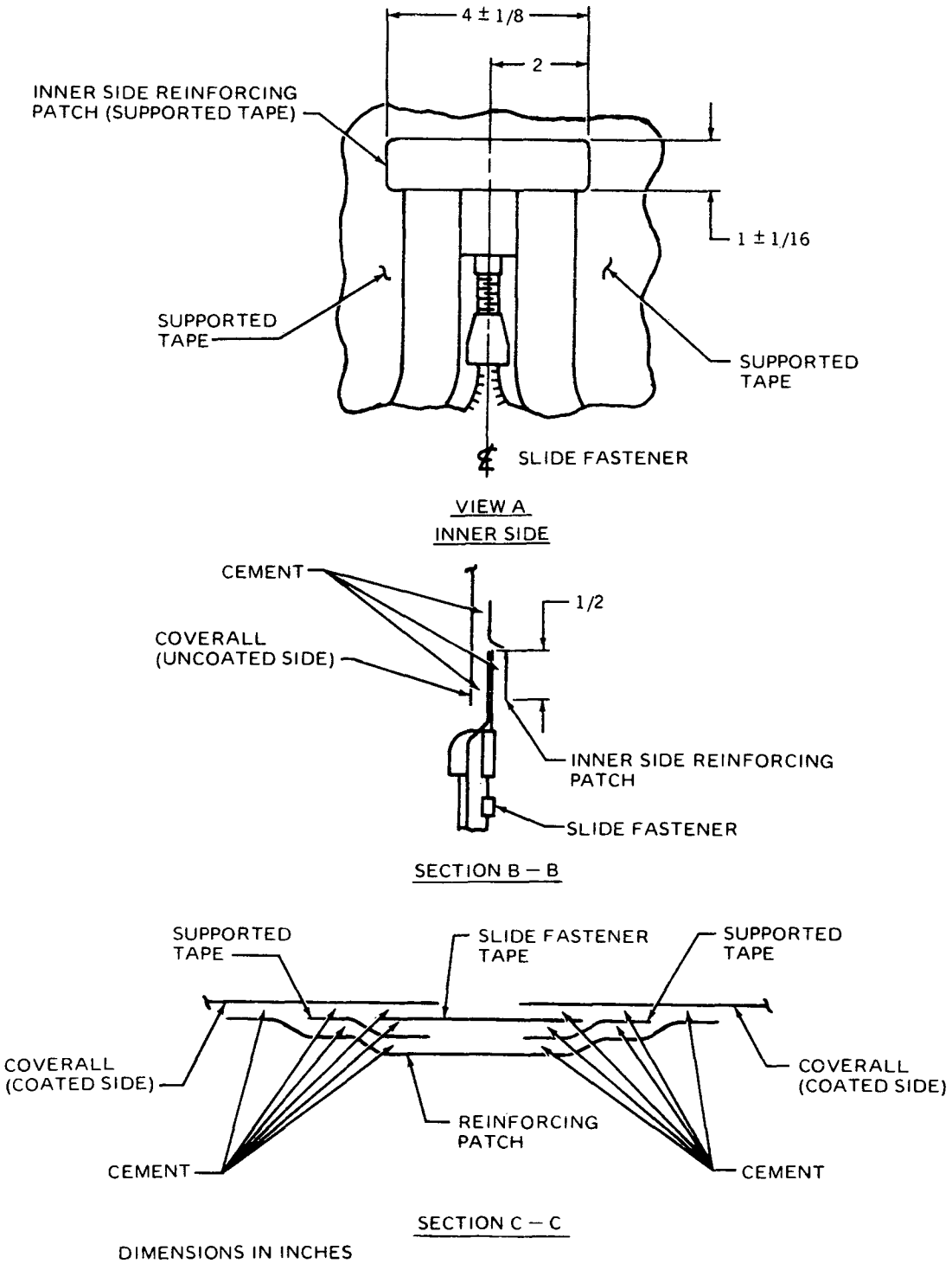



FIGURE 4. COVERALL SECTIONS



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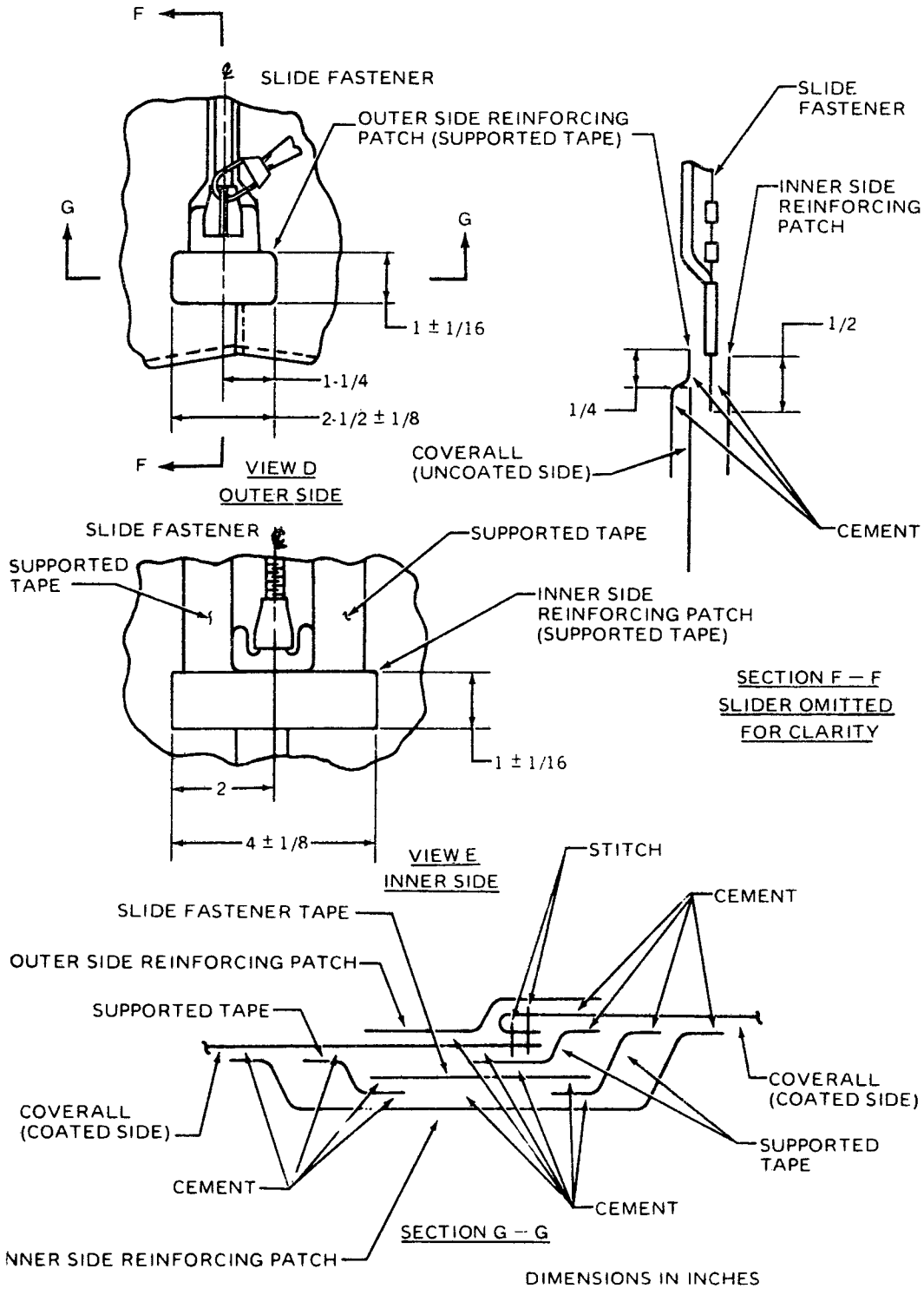



FIGURE 5. COVERALL SECTIONS



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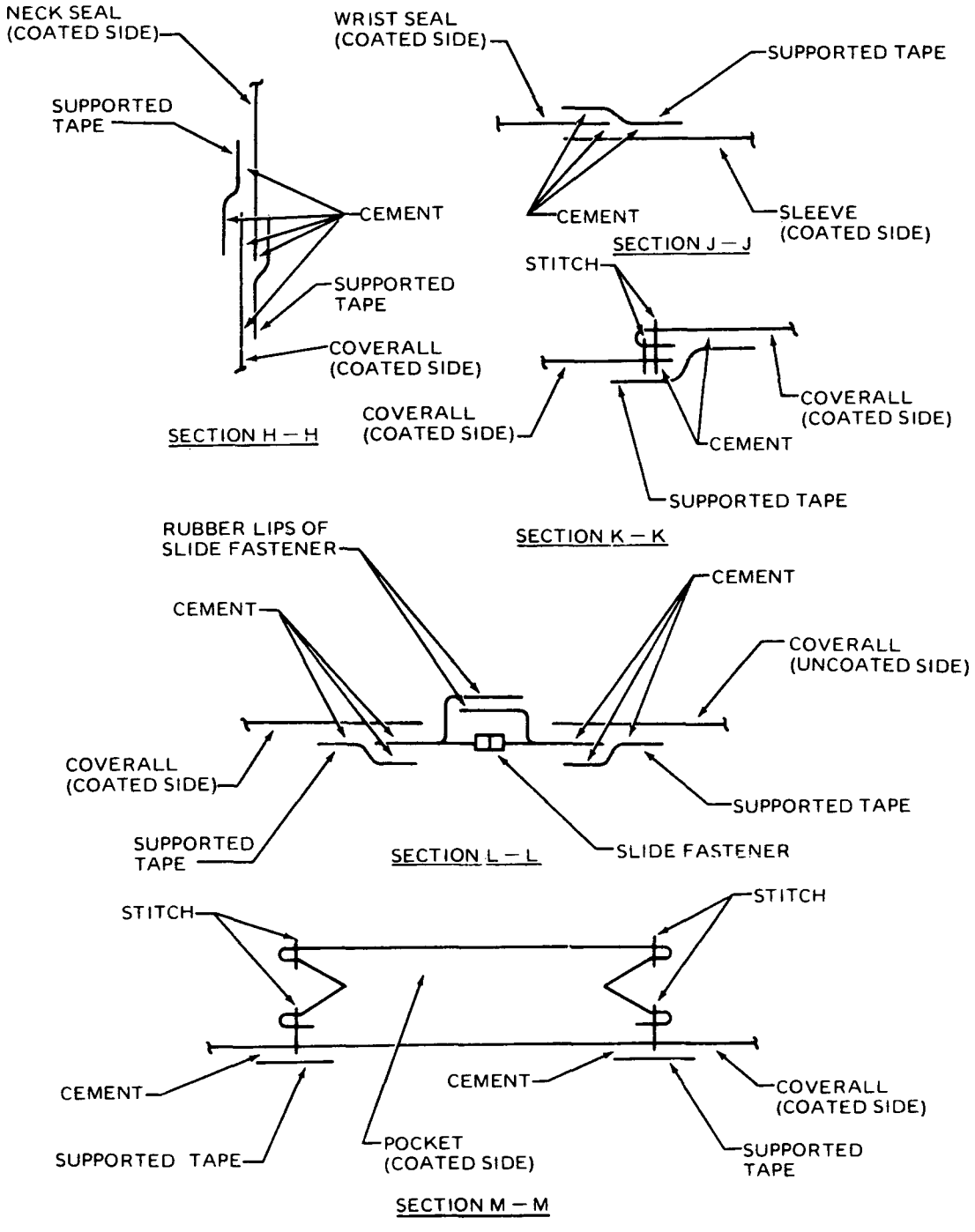
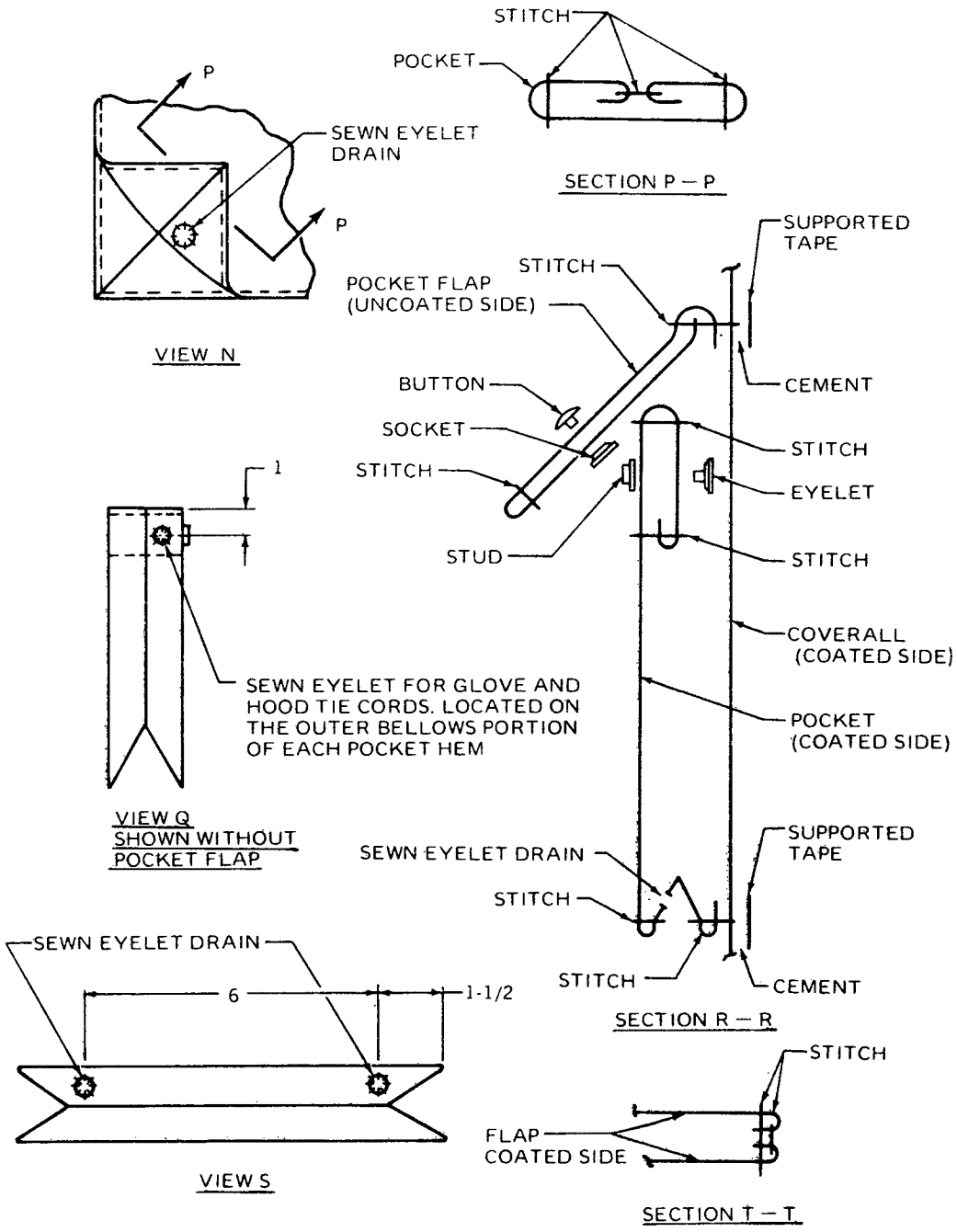



FIGURE 6. COVERALL AND POCKET SECTIONS



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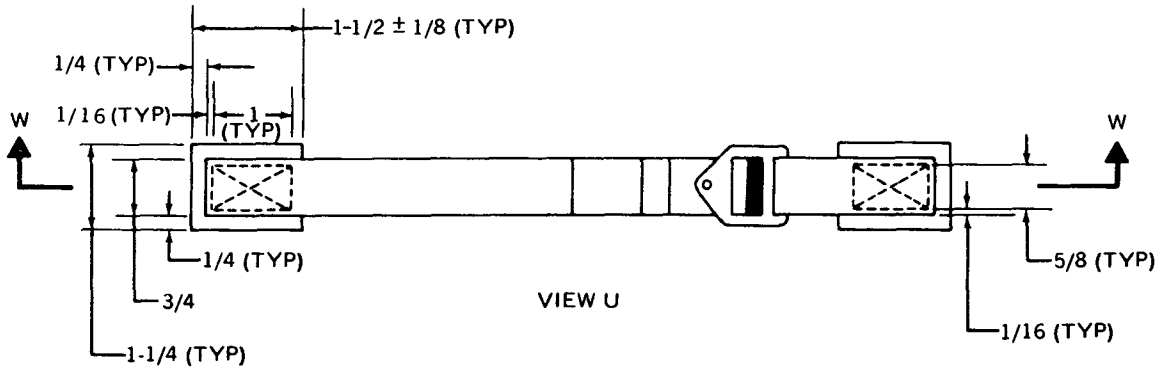

DIMENSIONS IN INCHES

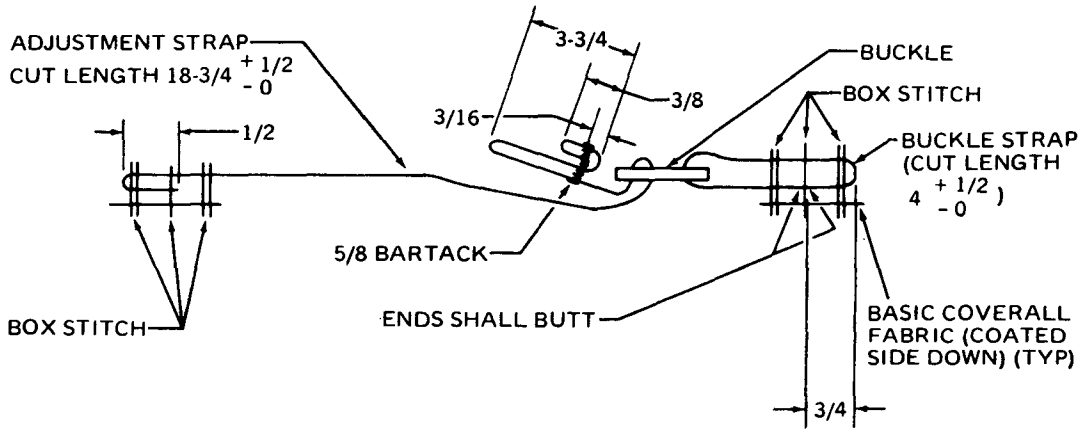
TOLERANCES ± 1/8

FIGURE 7. POCKET SECTIONS



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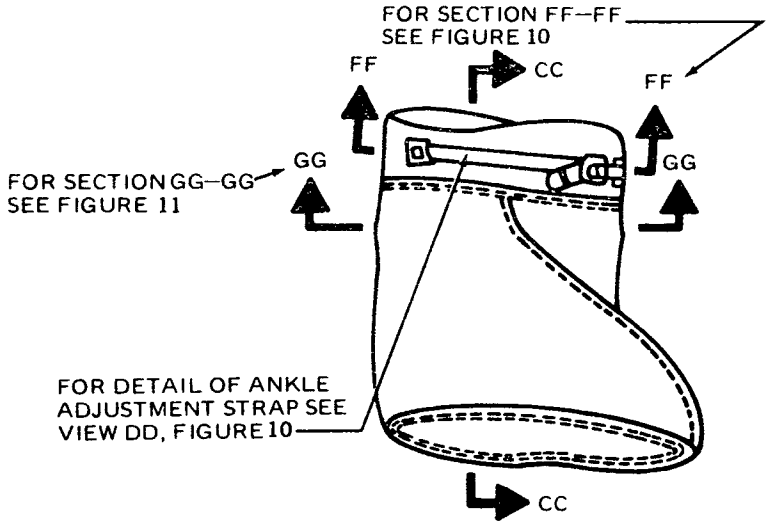

SECTION W-W

NOTES:
1. TWO WAIST ADJUSTMENT STRAPS SHALL BE CEMENTED TO COVERALL.
2. ALL DIMENSIONS ARE IN INCHES.

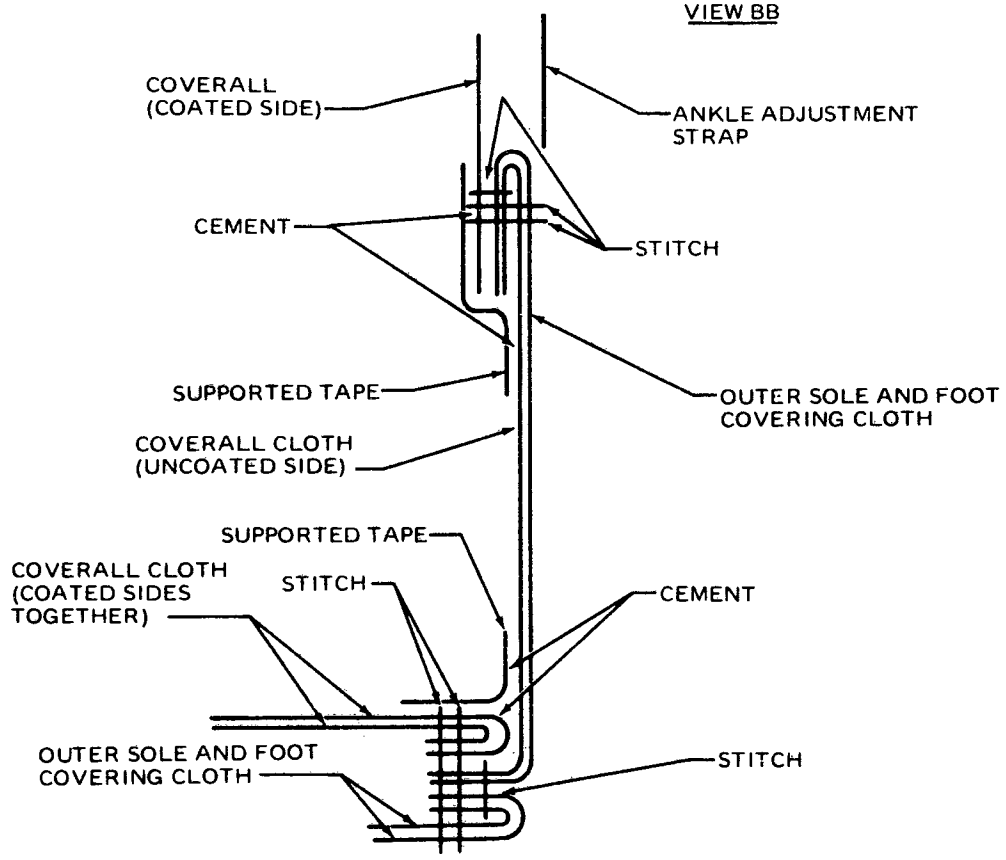
FIGURE 8. WAIST ADJUSTMENT STRAP



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VIEW BB

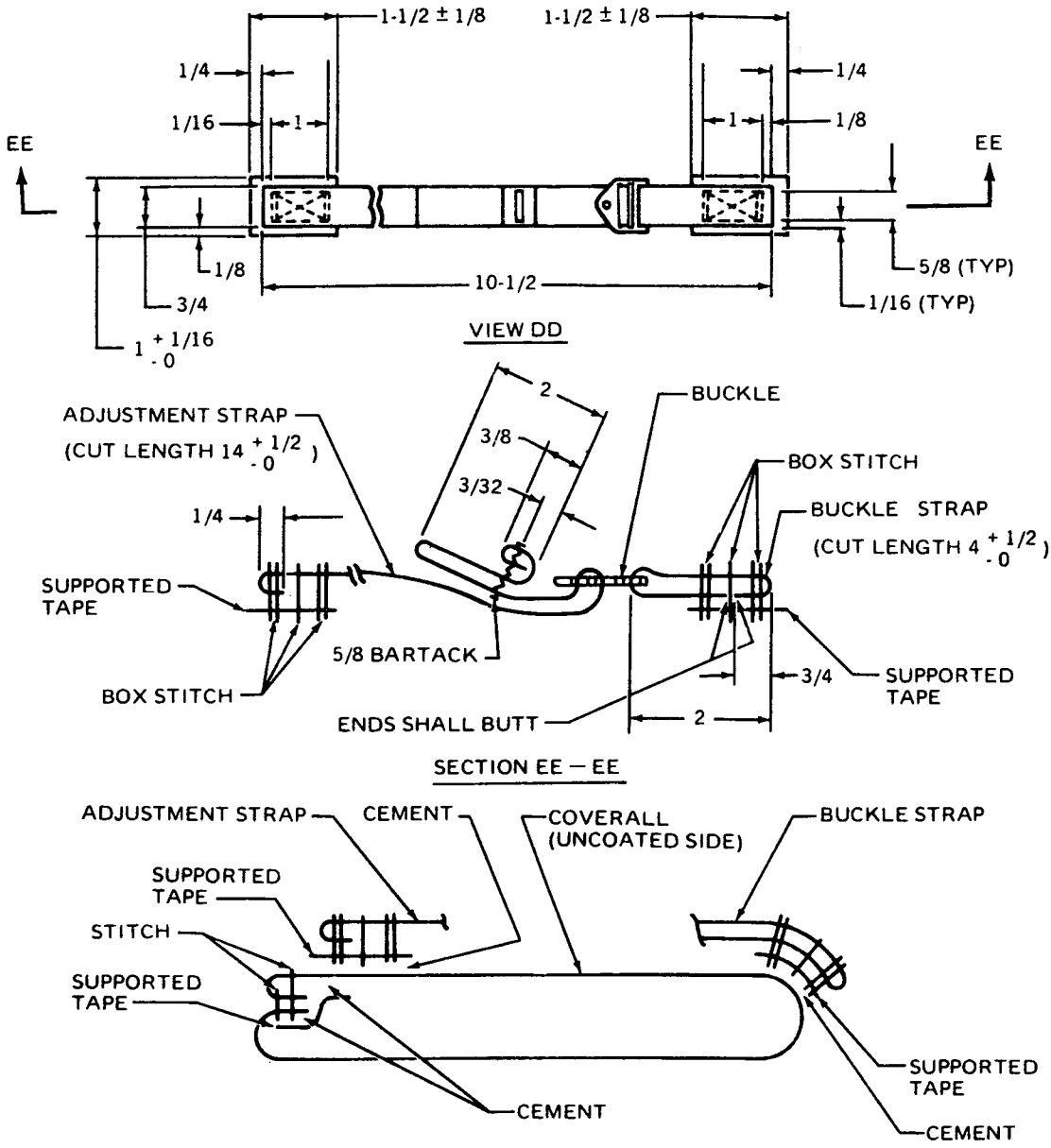



SECTION CC—CC

FIGURE 9. CONSTRUCTION DETAILS FOR BOOT



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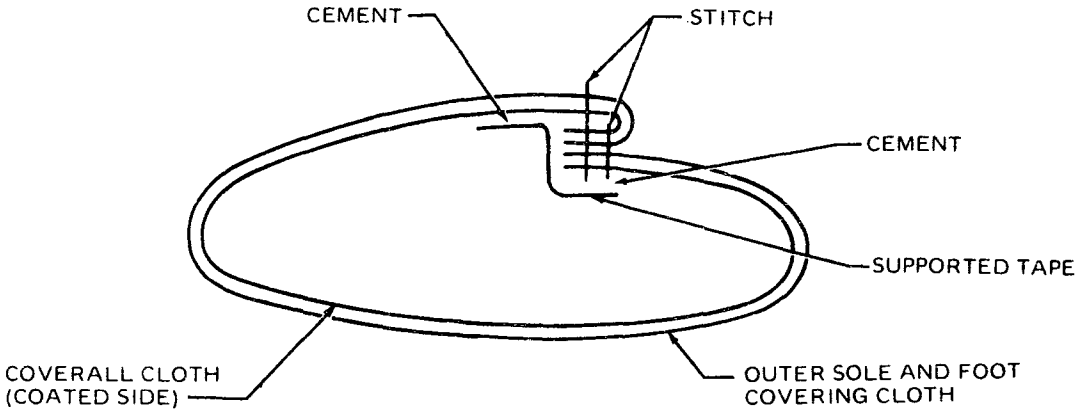

DIMENSIONS IN INCHES, UNLESS OTHERWISE SPECIFIED, TOLERANCES ON DIMENSIONS SHALL BE AS FOLLOWS:

| DIMENSIONS | TOLERANCES |
|---|---|
| 0 TO 1/2 INCL | ± 1/32 |
| OVER 1/2 to 1 INCL | ± 1/16 |
| OVER 1 TO 5 INCL | ± 1/8 |
| OVER 5 TO 11 INCL | ± 1/4 |

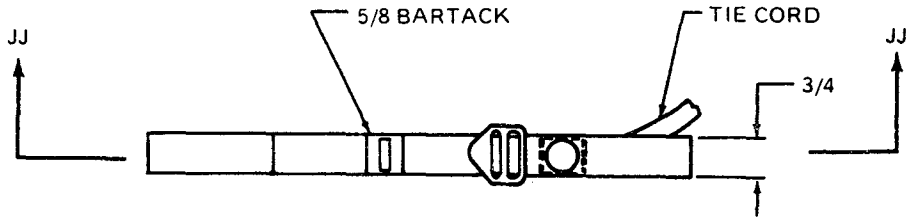
FIGURE 10. BOOT SECTIONS



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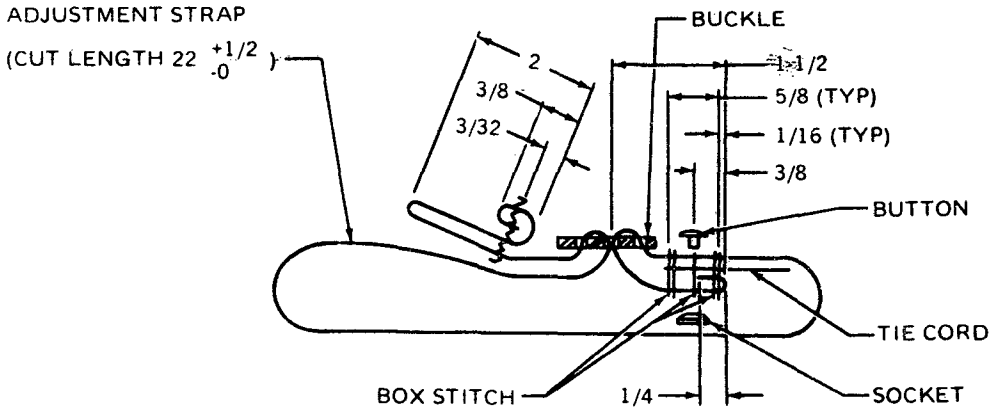



SECTION GG – GG
(BOOT SECTIONS)




VIEW HH

GLOVE SECTIONS




DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED
TOLERANCES SHALL BE AS
SPECIFIED IN FIGURE 10.

SECTION JJ – JJ

(GLOVE SECTIONS)

FIGURE 11. BOOT AND GLOVE SECTIONS



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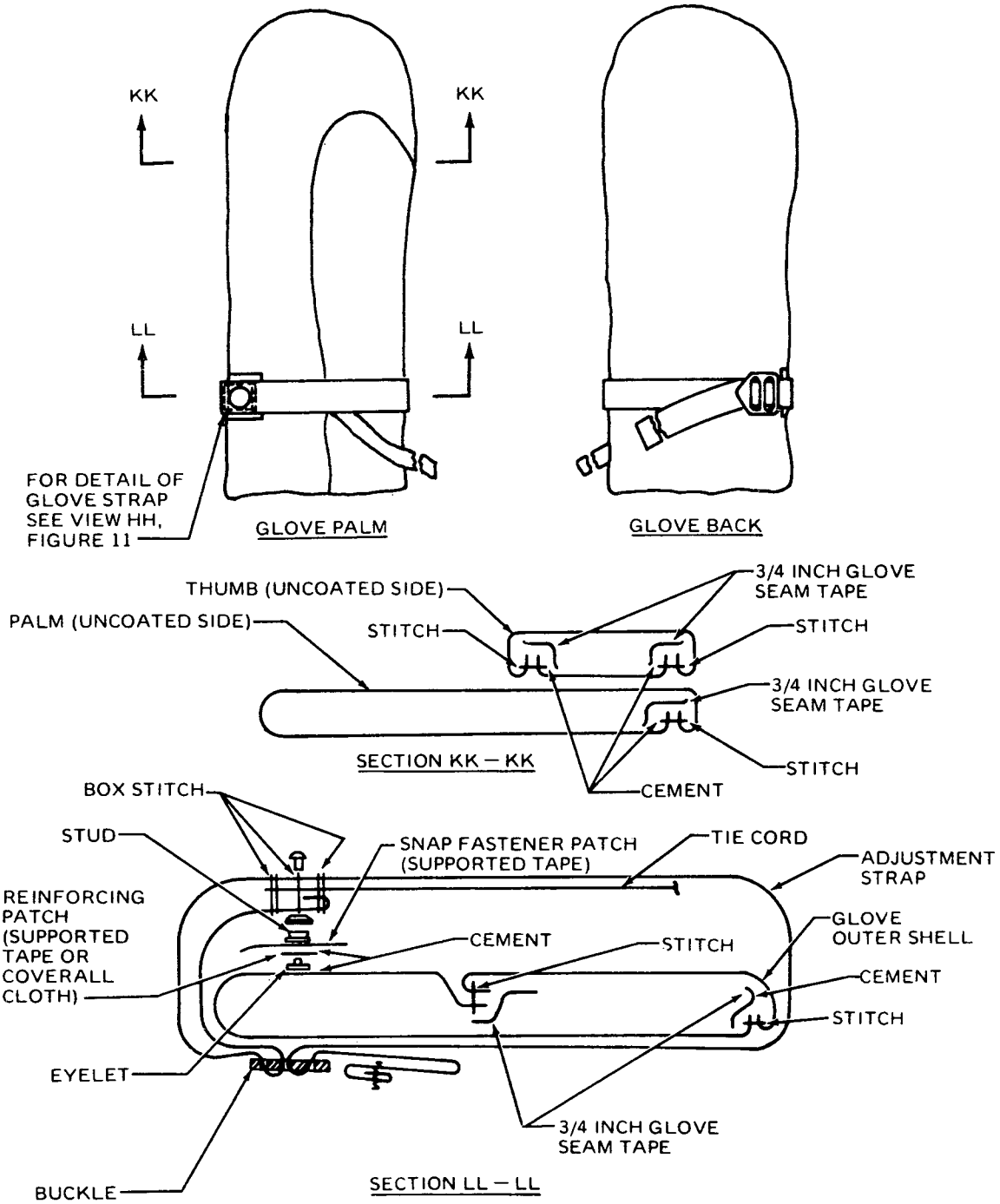
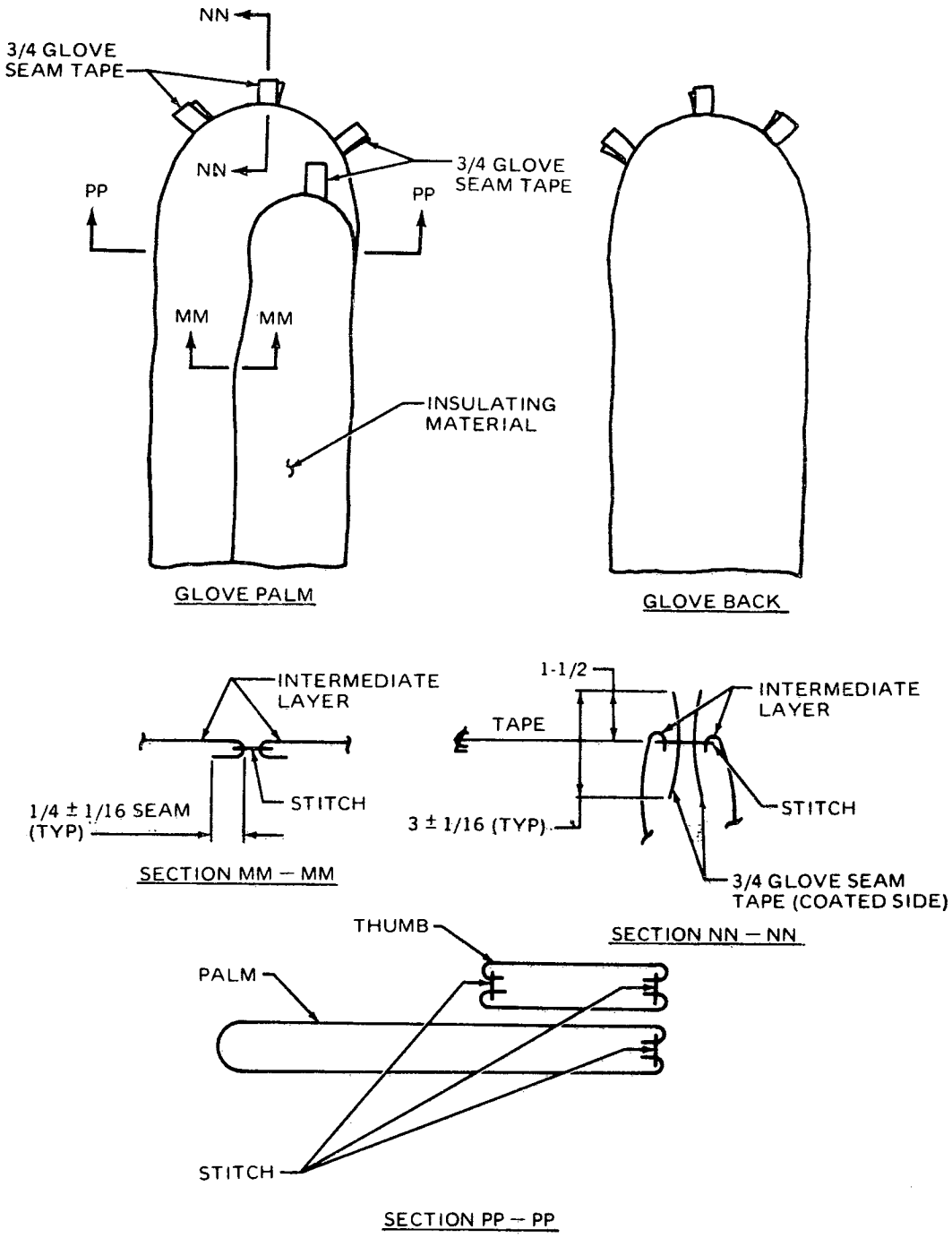



FIGURE 12. GLOVE SECTIONS (OUTER SHELL)



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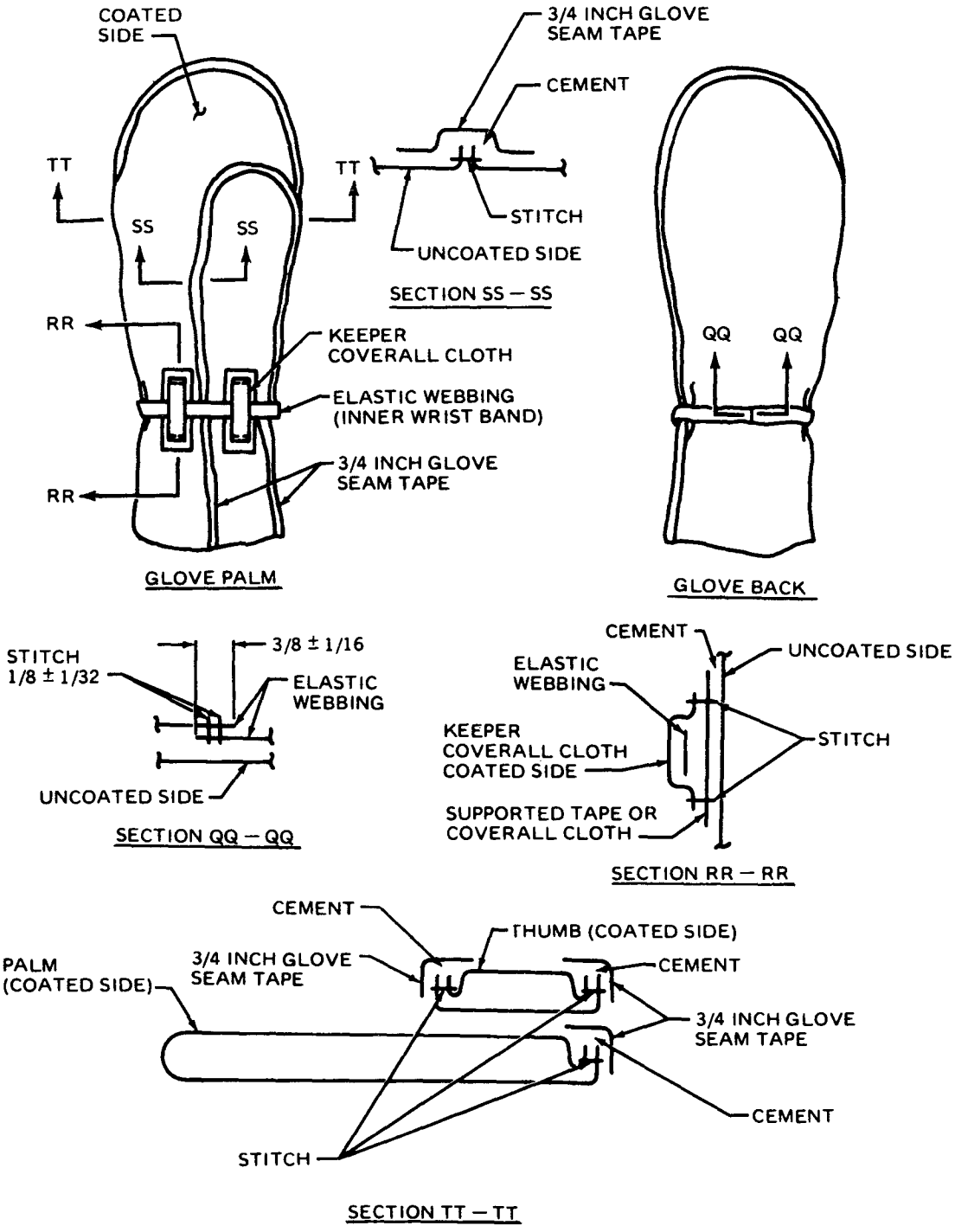



DIMENSIONS IN INCHES

FIGURE 13. GLOVE SECTIONS (INTERMEDIATE LAYER)



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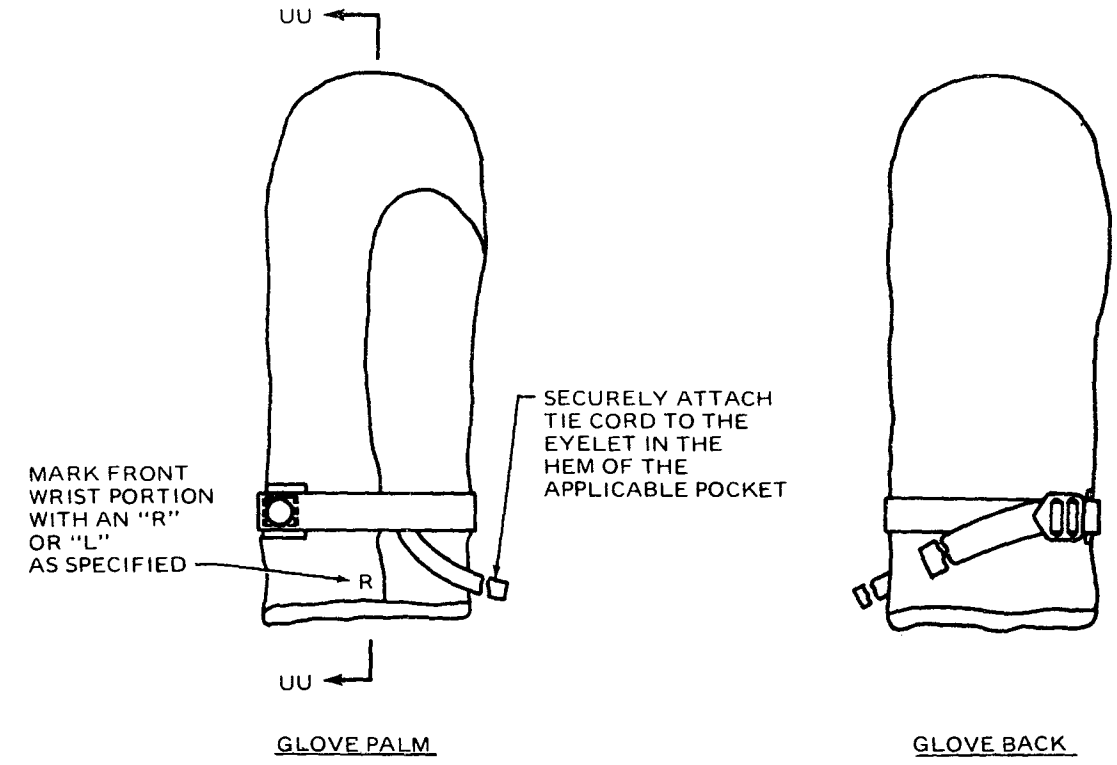


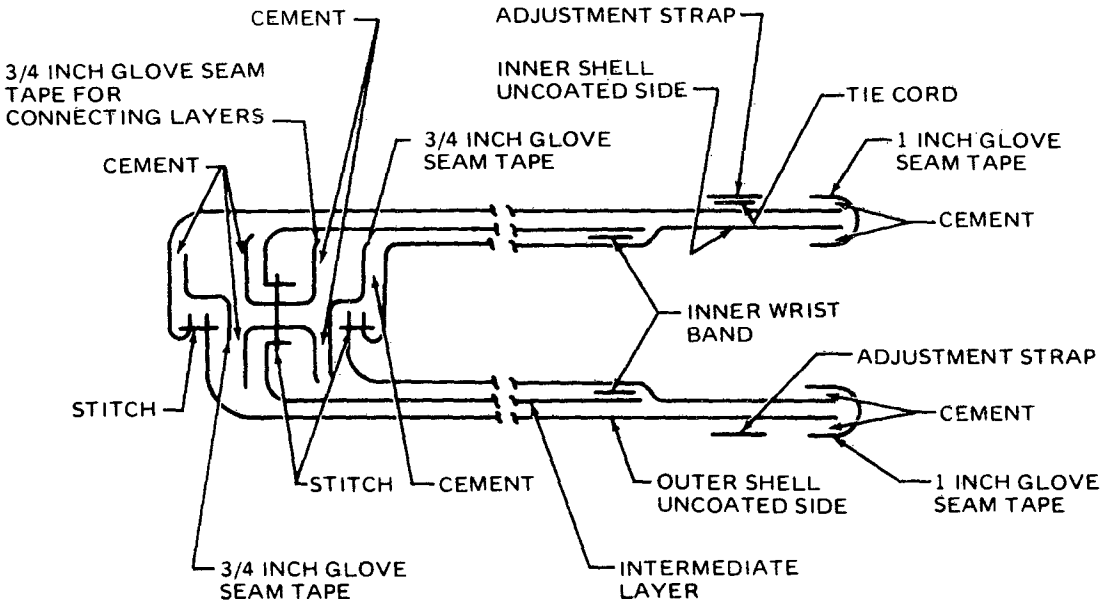

DIMENSIONS IN INCHES

FIGURE 14. GLOVE SECTIONS (INNER SHELL)



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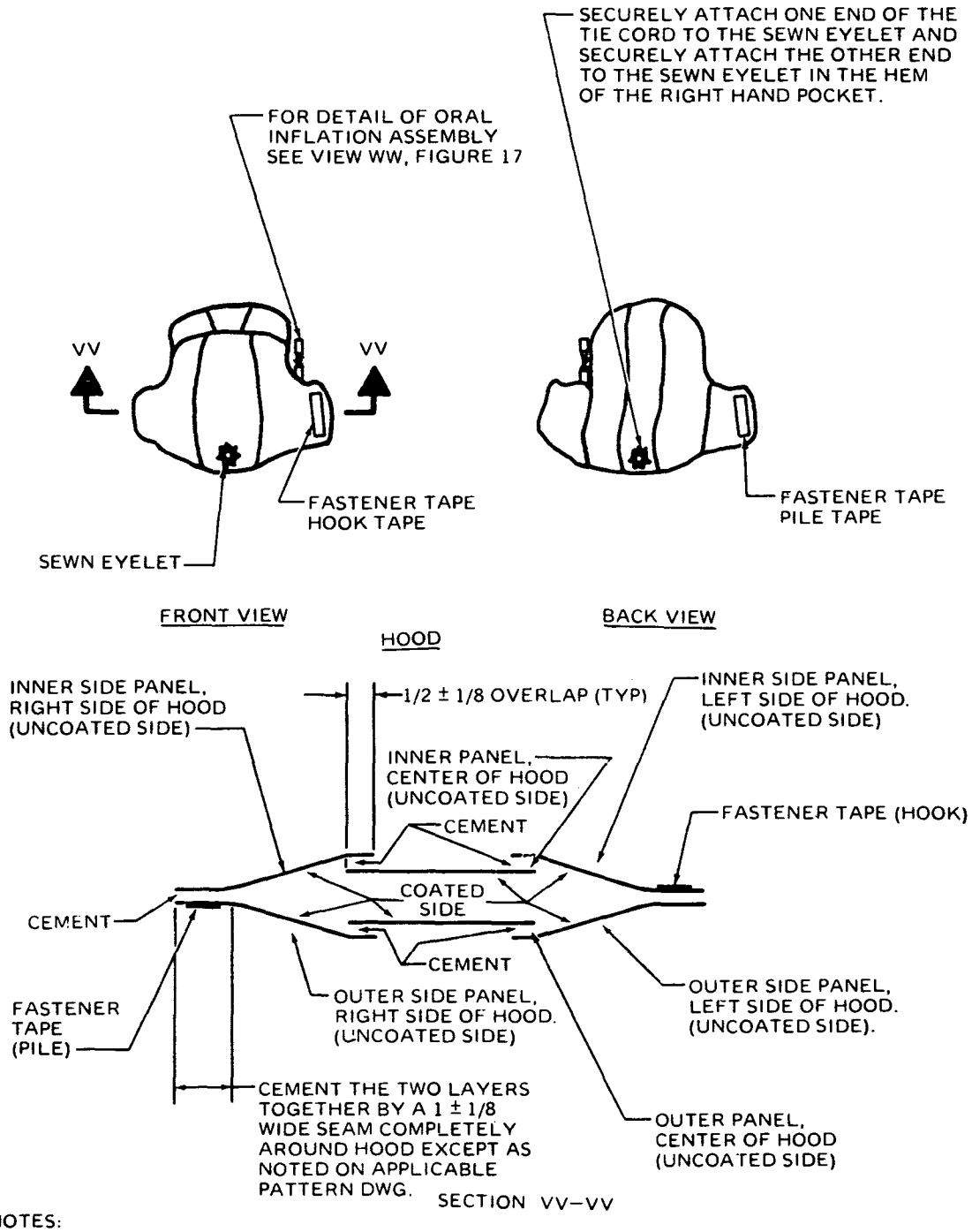




SECTION UU – UU

FIGURE 15. GLOVE SECTIONS (ASSEMBLY)



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NOTES:
1. STITCH FASTENER TAPES TO RESPECTIVE PANELS 1/16 ± 1/32 FROM OUTER EDGE OF TAPE ALL AROUND.
2. STITCHING SHALL BE PERFORMED PRIOR TO LAP SEAMING.
3. DIMENSIONS ARE IN INCHES.

FIGURE 16. HOOD SECTIONS



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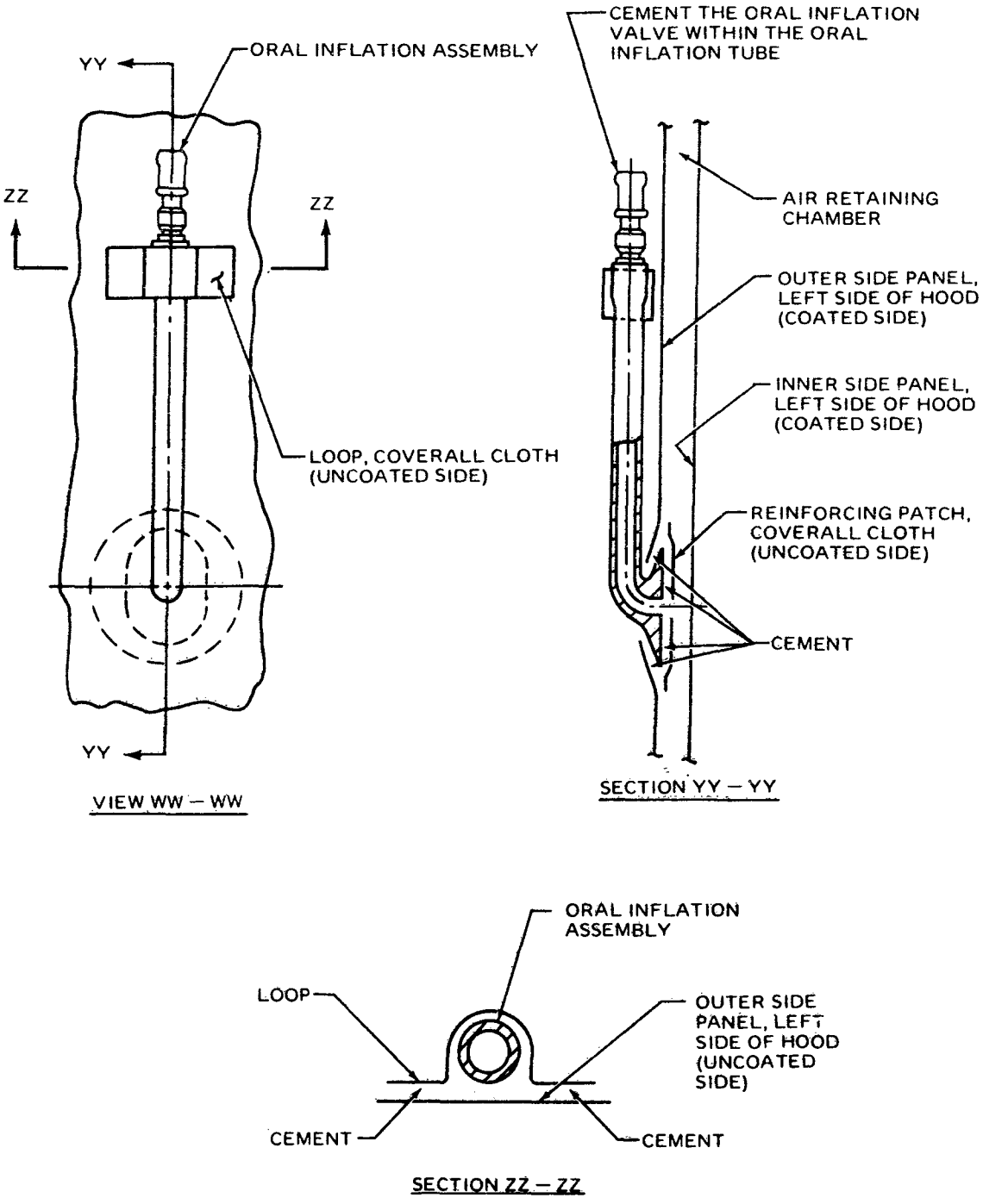



FIGURE 17. HOOD SECTIONS



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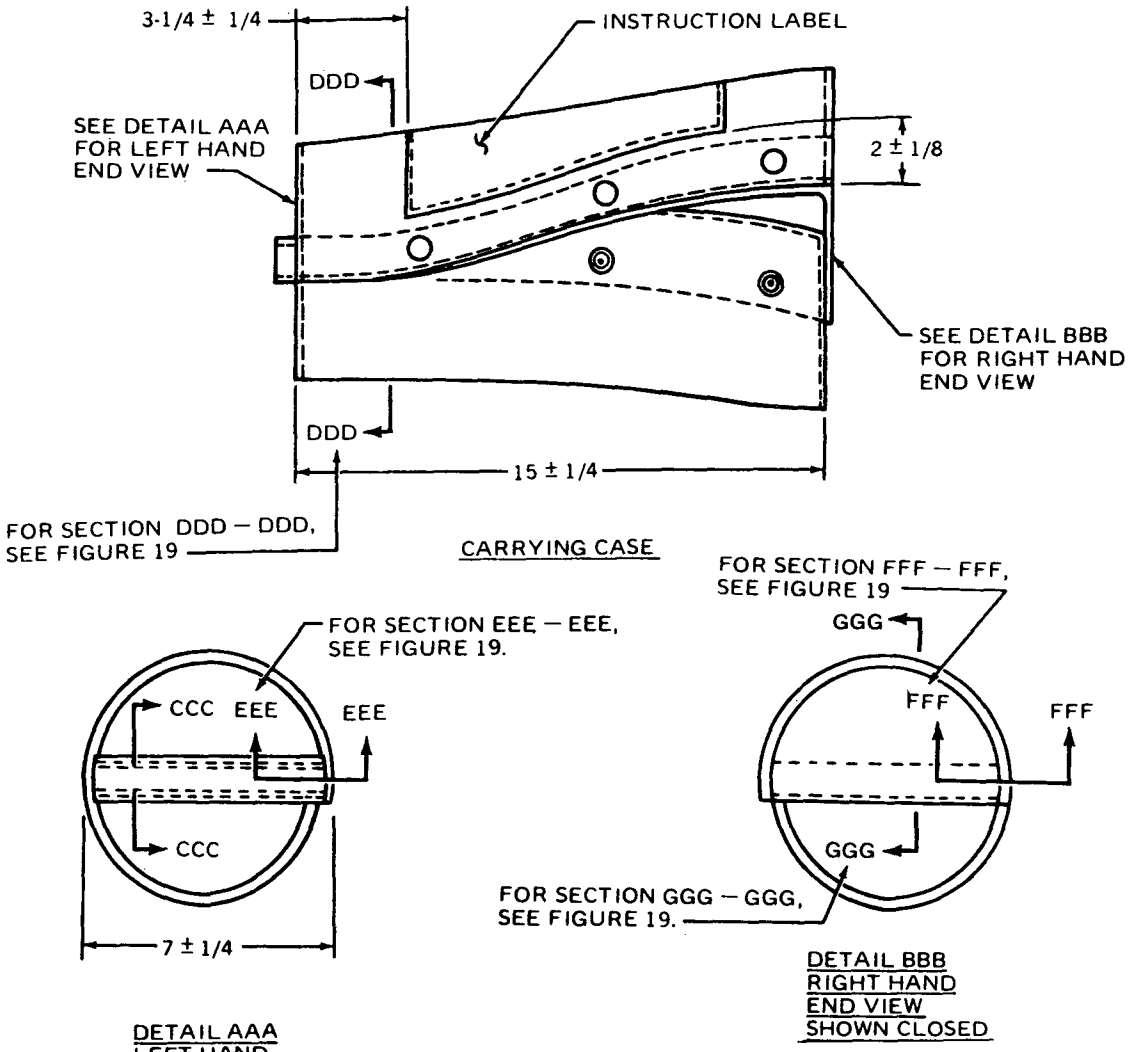
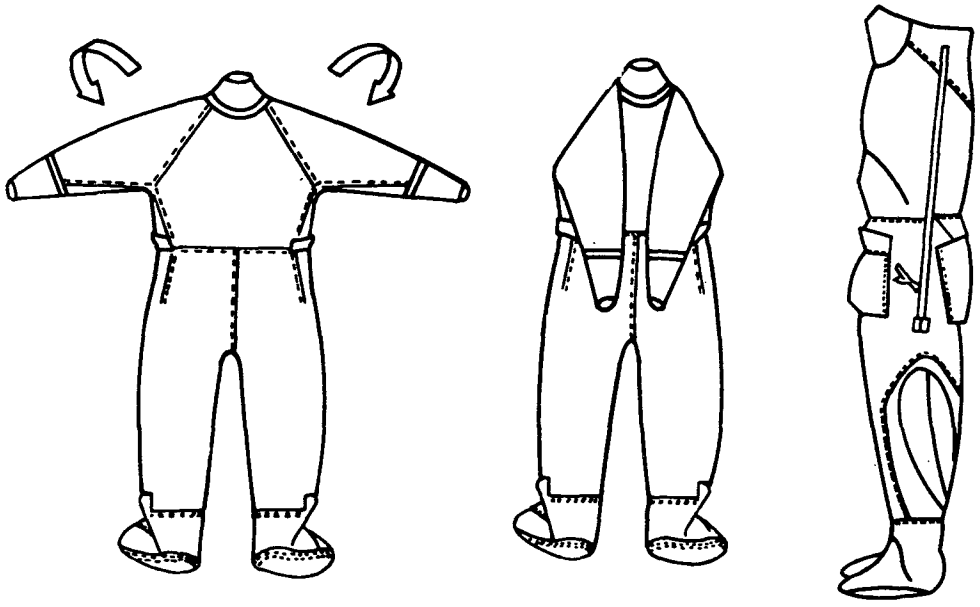



FIGURE 18. CARRYING CASE SECTIONS



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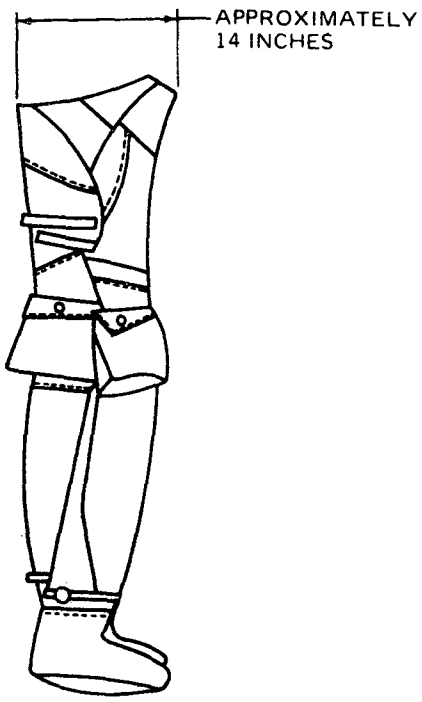


STEP NO. 1 STEP NO. 2 COMPLETED STEP NO. 3

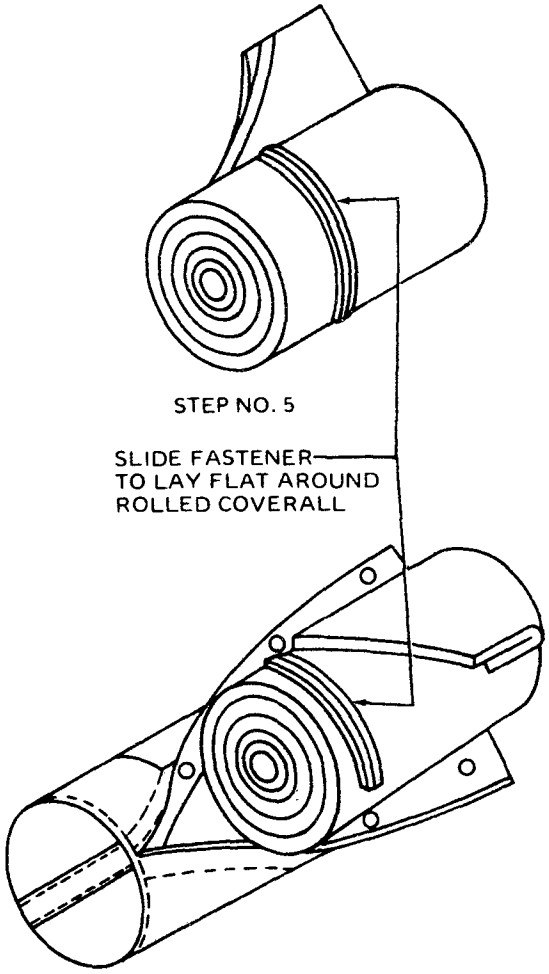
FIGURE 20. ASSEMBLING PROCEDURES (STEP NO. 1 THROUGH 3)



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STEP NO. 4




STEP NO. 5

STEP NO. 6

FIGURE 21. ASSEMBLING PROCEDURES (STEP NO. 4 THROUGH 6)



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

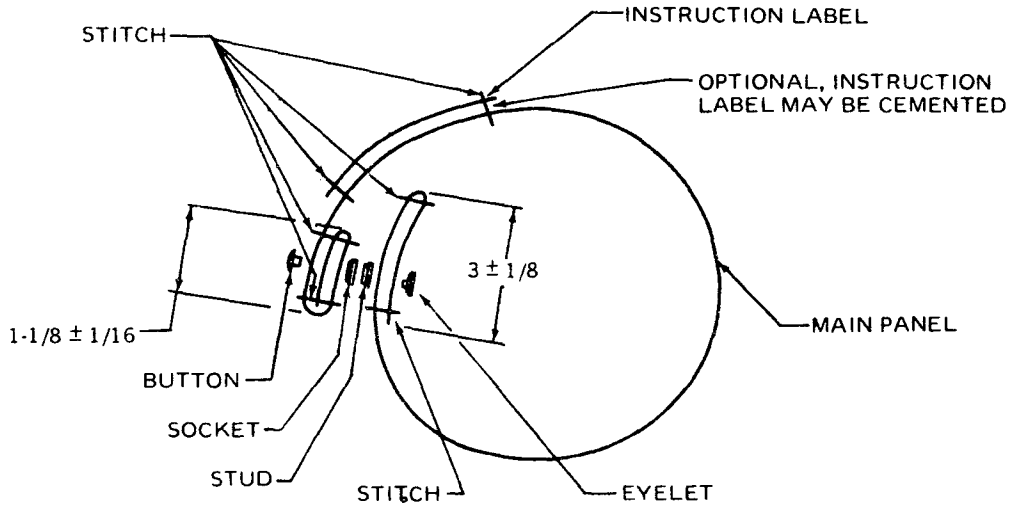
☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

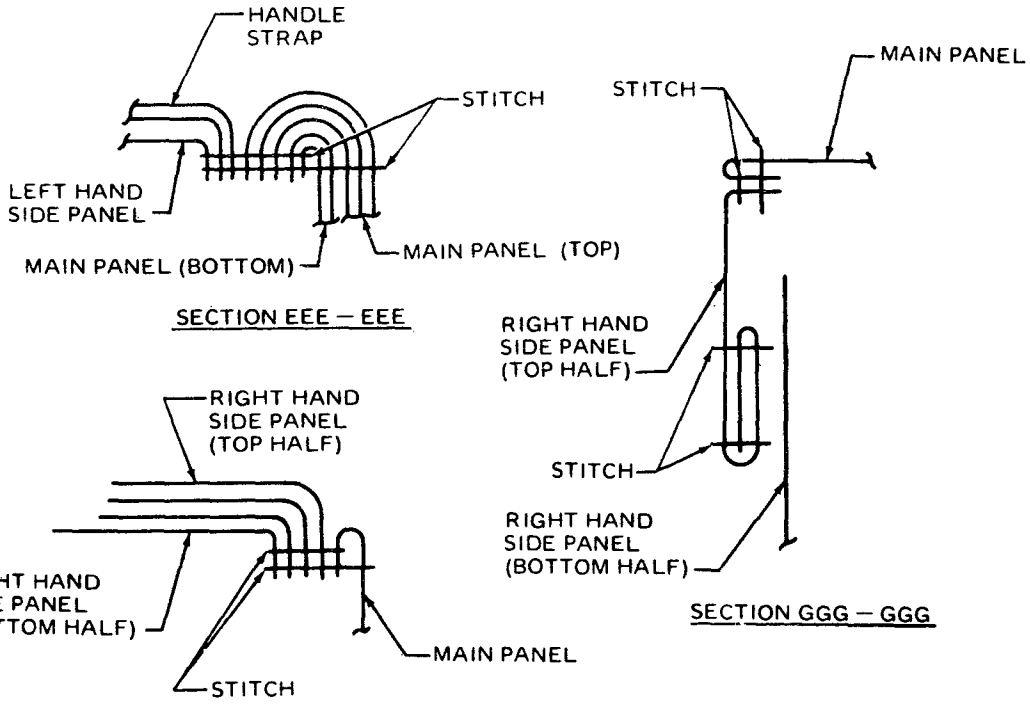
SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD. FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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SECTION DDD – DDD




SECTION EEE – EEE

SECTION GGG – GGG

SECTION FFF – FFF

DIMENSIONS IN INCHES

FIGURE 19. CARRYING CASE SECTIONS



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